

Sum-Score Versus Latent Variable Approaches to Growth Curve Modeling: Scale Alignment and Parameter Divergence under Measurement Violations

by

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Abstract

Abstract goes here.

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Chapter 1 Introduction

1.1 Sum Scores vs. Latent Variables: A Persistent Methodological Divide

In quantitative studies in education and psychology, researchers rely on specific scales to measure constructs of interest that will subsequently be tested in statistical models. Historically, composite scores (e.g., sum scores or mean scores) have been widely used due to their simplicity and interpretability, supported by a deep foundation in classical test theory where sum scores are central to estimating internal consistency. Because composite scores are computationally simple and their derivation is transparent, they are highly favorable to study replication and interdisciplinary communication. However, while composite scores, particularly sum scores, are convenient for direct application in fields such as clinical psychology, their validity is often questioned, especially when comprehensive evidence of construct validity is not fully disclosed.

A latent variable model, specifically within the framework of factor analysis and broader structural equation modeling (SEM), is widely regarded as a rigorous alternative to the composite score. Its primary advantage lies in explicitly modeling the measurement process by incorporating residual terms (i.e., measurement error). A latent variable model relates observed indicators to an underlying reflective construct via a specified factor structure. Because it is unrealistic in empirical research to assume that measurements are obtained without error, SEM treats responses from multiple indicators as a manifestation of this latent construct. Furthermore, because a single indicator rarely captures a complex concept adequately, utilizing latent factors provides significant conceptual advantages alongside supplementary validity information. Nevertheless, latent variable models are inherently more complex and require estimating more parameters than composite score approaches, which assume error-free measurement. Additionally, they require arbitrary identification constraints to establish the latent metric, which can be counterintuitive and present a barrier to communicating results to applied audiences.

The choice between composite scores and latent variables has been a concern for long not only among applied researchers but also among quantitative methodologists. Recently, there has been active debate regarding under which scores are acceptable, and when the two approaches yield convergent or divergent conclusions. For consistency and to align with the primary focus of recent methodological literature, this paper will narrow its focus from general composite scores to the "sum score" moving forward.

For a sum score to adequately approximate a latent variable, unidimensionality is a prerequisite. When unidimensionality is violated, the use of sum scores becomes theoretically imprecise and statistically problematic. The dimensionality must be evaluated using a factor-analytic approach, not merely inferred from the reliability of the sum score. A further requirement is that the data must satisfy at least the assumption of an essentially tau-equivalent model. Unidimensionality alone does not guarantee essential tau-equivalence (i.e., equal factor loadings across all indicators).

It should be noted, however, that tau-equivalence itself does not promise an acceptable underlying dimensionality. In practice, researchers must verify an "acceptable" level of dimensionality through comprehensive evaluations, including the magnitude of standardized factor loadings (e.g., $> .70$) and sometimes comparisons against competing models (e.g., bifactor models or exploratory factor analysis). When an acceptable level of unidimensionality is established and the stricter conditions of a parallel model are met, the parameter estimates derived from both approaches will be almost identical.

1.2 Why Longitudinal Settings Complicate the Choice

In cross-sectional settings, the conditions under which sum scores and latent variables yield equivalent results are relatively well understood. However, when the research question shifts from measuring a construct at a single occasion to the multiple time points, the relationship between these two approaches becomes more complicated.

Behavioral scientists are often interested in the growth of latent variables that can only

be indirectly measured through manifest indicators. Latent growth curve modeling is one of the most widely used models for capturing growth trajectory over time. In a first-order LGCM (1-LGCM), sum scores (or other composite scores) at each time point serve as observed variables. In a second-order LGCM (2-LGCM), a measurement model is specified with indicators measured repeatedly, and growth factors are built on top of the resulting latent factors.

A natural question is whether the 1-LGCM and 2-LGCM can recover the same growth parameters or not. Under specific conditions, they do. When strict measurement invariance holds and an identification method known as effect coding is applied, the sum score at each time point turns out to be an equivalence to the effect-coded latent factor with measurement noise. That is, the 1-LGCM fitted to sum scores approximately recovers the same growth factor means and covariances that the 2-LGCM estimates directly.

The conditions required for interchangeable use of sum-score and latent variable are more demanding for longitudinal setting as longitudinal data introduce at least two systematic threats that can break it. First, when the same indicators are observed repeatedly across multiple occasions, its residual at one wave tends to correlate with its residual at adjacent time points, known as correlated uniqueness. Second, the measurement parameters, such as factor loadings and intercepts, may not remain the same over time, referred to as measurement non-invariance. Both of these threats are common in applied research, and both have the potential to cause the 1-LGCM and 2-LGCM to diverge in their growth parameter estimates.

In a 2-LGCM, correlated uniquenesses can be modeled directly by specifying residual covariances within same item over time. The growth parameters are then estimated after having separated this from true construct change. A sum-score 1-LGCM has no mechanism for this separation. The omitted correlated uniqueness may systematically bias the estimated parameters in 1-LGCM. Whether this bias is large enough to matter in practice is the question needs to be addressed.

The second threat of measurement non-invariance is problematic not only because it changes the meaning of construct, but also it changes the scale fundamentally. When loadings or intercepts drift over time, these sums become time-varying as well, and the sum-score is no longer on a stable metric. Suppose, for example, that loadings and intercepts monotonically increase across waves; in this case, an identical unit of latent change yields a spuriously larger change in the observed sum score at later occasions. Unlike correlated uniqueness, which primarily distorts variance and covariance parameters, measurement non-invariance can confound the mean structure as well, the estimated growth rate itself, not just individual differences around it.

1.2.1 A Gap in the Literature

The existing literature has not addressed these two threats within a unified comparison in LGCM framework. Although previous studies have examined the effects of measurement non-invariance and correlated uniqueness on growth parameter estimation, they have typically done so within either a composite or a latent variable framework, rather than across both simultaneously. A primary reason for the scarcity of direct comparisons between 1-LGCM and 2-LGCM is the fundamental lack of metric equivalence between their scales. Addressing these methodological gaps motivated the present research.

1.3 Purpose, Contributions, and Dissertation Overview

The primary purpose of this dissertation is to quantify the extent and direction of divergence in growth parameter estimates between the 1-LGCM using sum-scores and the 2-LGCM under conditions of omitted residual structure (CU) and measurement non-invariance (MNI). The investigation is conducted through two Monte Carlo simulation studies. Study 1 examines the isolated impact of CU while maintaining strict measurement invariance. Study 2 introduces MNI by incorporating drifts in factor loadings and item intercepts. To avoid confounding from raw scale differences, Study 1 applied effect-coding

identification to align the scales, and Study 2 evaluated the outcomes using scale-free metrics.

Chapter 2 reviews the literature on the debates surrounding sum scores and latent variables in general, and expands the discourse to the longitudinal data analysis context. In particular, to ensure the estimates from both models are compared on the same metric, Chapter 3 provides a detailed mathematical derivation for aligning the latent scale of the 2-LGCM with that of the sum-score 1-LGCM using the effect-coding identification method. Detailed research questions are presented in Chapter 4. Chapter 5 presents the method, results, and discussion for Study 1, and Chapter 6 does the same for Study 2.

This research makes four main contributions to longitudinal methodology. First, it systematically compares sum-score and latent modeling approaches under realistic data conditions frequently encountered in applied settings. Second, by establishing the mathematical transformation for scale alignment between the 2-LGCM and the sum-score 1-LGCM, this work provides a theoretical foundation for comparing these two distinct scoring frameworks. Third, it provides a quantitative assessment of the independent effect of CU in the context of growth model. Fourth, it demonstrates how MNI interacts with the implicit metric of sum scores, producing predictable patterns of bias. Ultimately, these findings offer practical guidance for researchers when choosing between sum scores and latent variables for longitudinal data analysis.

Chapter 2 Literature Review

2.1 Methodological Review of Scoring Methods

For many years and especially in recent discussions, scholars have raised concerns about score reporting practices, emphasizing that it is critical for open science to share the exact way measures are used (e.g., number of items, reliability, validity evidence), because these choices directly affect replicability and reproducibility of study (Barry et al., 2014; Flake et al., 2022; Hogan and Agnello, 2004; Lawson et al., 2025). However, as Flake et al. (2022) demonstrated, it is still common to report little about crucial information on validity of the scales being used. This concern aligns with recent debates on composite scores, mainly sum scores, often without thorough validation of the scores (McNeish and Wolf, 2020) as opposed to the arguments in favor of composite scores (Widaman and Revelle, 2022). For the current dissertation, I will focus on the sum score as one of several types of composite scores (e.g., mean scores).

2.1.1 Historical Backgrounds

Classical Test Theory Since its development in the early 20th century, Classical Test Theory (CTT) has remained a foundational measurement framework widely utilized by psychometricians (Lord and Novick, 1968; Sijtsma et al., 2024a). Because the debate between sum scores and latent variables is rooted in this framework, it serves as the starting point for this study. CTT decomposes observed scores (X) into true score (T) and errors (E) as: $X = T + E$. Following approach proposed by Sijtsma et al. (2024a), CTT further assumes that: (1) errors and true scores from different tests are uncorrelated, $\sigma(E_j, T_k) = 0$; and (2) error terms from different tests are uncorrelated, $\sigma(E_j, E_k) = 0$. Because CTT relies on these relatively lenient assumptions regarding error terms, the model does not strictly define the nature of the true score beyond the basic equation. Consequently, some argue that CTT is effectively a “non-dimensional” error model

(Sijtsma et al., 2024b).

Here, reliability is defined as proportion of variance in observed X explained by variance in T the true score:

$$\rho = \frac{\text{Var}(T)}{\text{Var}(X)} = 1 - \frac{\text{Var}(E)}{\text{Var}(X)}. \quad (1)$$

Although error variance is not directly measurable, reliability can be approximated through internal consistency. Coefficient α , introduced by Guttman (1945) as λ_3 and later popularized and interpreted by Cronbach (1951), is formulated as:

$$\alpha = \frac{k}{k-1} \left(1 - \frac{\sum_{i=1}^k \text{Var}(X_i)}{\text{Var}(\text{Sum}_i)} \right), \quad (2)$$

where the sum score is defined as the total of all k items. Intuitively, $\sum_{i=1}^k \text{Var}(X_i)$ captures sum of item-specific variability, whereas $\text{Var}(\text{Sum}_i)$ reflects shared variance (covariance) among items. While α is widely used as a measure of internal consistency, it is important to note its utility and interpretation have been debated that may not adequately represent the reliability of a test (Sijtsma, 2009). Historically, while the foundations of CTT preceded coefficient α , the two have long been used together due to their clear conceptual connection (Miller, 1995). This is particularly true for sum scores, as they apply to situations where internal consistency is a primary measurement concern.

Factor Analysis and Latent Variable One caveat regarding reliability is that internal consistency does not necessarily guarantee construct validity (Cook and Beckman, 2006; DeVellis, 2006). To evaluate the internal structure and assess whether the items does measure a single construct of interest, a latent-variable factor-analytic model should be fitted (Thompson and Daniel, 1996). Before proceeding, it is important to clarify the measurement specification whether indicators are formative or reflective to avoid confusion. As Coltman et al. (2008) noted, the nature of the latent construct determines this choice: if the construct exists first and causes variation in the indicators, the measurement is

reflective; if the construct is formed as a combination of indicators, it is formative. This paper focuses exclusively on the reflective case, in which the latent construct causes variation in the manifest indicators. Accordingly, I do not address weighted composite scores used to measure formative constructs (e.g., socio-economic indices derived via principal component analysis; Mazziotta & Pareto, 2024).

Latent-variable factor analysis, rooted in Jöreskog (1969), serves as the foundation for covariance-structure modeling and Structural Equation Modeling (SEM). In this framework, an observed score Y_i is caused by a latent factor F :

$$Y_i = \tau_i + \lambda_i * F + z_i, \quad (3)$$

where i indexes each item, τ is intercept, λ is the item-specific factor loading or weights, F is the latent factor score and z is the residual. As one of evidences for the intended construct, standardized loadings are expected to exceed 0.70 (Comrey and Lee, 2013), and the one-factor model should fit the data adequately by common cut-off of approximate fit-index cutoffs (e.g., Hu & Bentler, 1999). Unlike the sum-score approach in CTT, this model represents measurement error explicitly through item-specific residual variances (measurement error) rather than as a single overall partition of observed-score variance.

2.1.2 Underappreciated Strengths of Sum Scores

Sum scores may appear “intuition driven” (Mislevy, 2024), but they often provide strong theoretical and practical benefits. Sijtsma et al. (2024a) clarified that under binary prediction, sum score’s stochastic ordering properties imply that sum score can preserve the ordering of latent trait well-the most evident case being the sum score as a sufficient statistic of the Rasch model (Andersen, 1977; Zwitser and Maris, 2016). When items are close to essentially tau equivalent (i.e., equal weights loadings across items) and the scale is unidimensional, a sum score serves as a practical summary (Widaman and Revelle, 2022)

and a reasonable proxy for the construct of interest. From a reliability standpoint, under essentially tau-equivalence, coefficient α equals the reliability not merely a lower bound (Sijtsma et al., 2024a). Thus, a sufficiently high α can be taken as supportive evidence consistent with unidimensionality implied by tau-equivalence. Practically, also when sample sizes are limited and latent score estimates can become unstable or noisy, sum scores can serve as a useful alternative to latent variable (Sijtsma et al., 2024a).

In a nutshell, throughout ongoing debates, two arguments are often emphasized by defenders of sum scores: (1) when scores are used for classification, screening, or one-to-one clinical settings, sum scores can be a practical proxy with little difference from latent-variable scores under limited sample sizes (Sijtsma et al., 2024a; Widaman and Revelle, 2022); and (2) sum scores may foster replicability of results because the raw-sum scoring rule is identical across users and studies, sometimes showing comparable or even better cross-validation performance than latent-variable scores that require arbitrary scaling choices (Sijtsma et al., 2024b; Widaman and Revelle, 2024).

2.1.3 The Demand for Measurement Precision

Despite the statistical and practical benefits of sum scores, their simplicity can become a serious liability when precise scale use is central. McNeish and Wolf (2020) reframed the sum score from a factor-analytic perspective, emphasizing that sum scoring implicitly imposes strict assumptions and criticizing the common practice of using sum scores without statistically validating those assumptions. In factor-analytic terms, summing across items implies a weighted composite; for a sum score, all weights are 1. This “equal unit weights” scoring rule can be linked to a restrictive measurement model, often described as a parallel model in which factor loadings, intercepts, and residual variances are equal across items (Graham, 2006). Widaman and Revelle (2022) showed that a perfect correlation between the sum score and the latent factor can be obtained under parallel-model assumptions, implying that these assumptions should be examined rather

than taken for granted. However, they clarified sum-score logic aligns more closely with essential tau-equivalence than with a strict parallel model.

Regarding the stochastic ordering properties of sum scores, although these properties can hold for binary items, they are criticized because many researchers in practice use ordinal responses (e.g., Likert-type scale) more frequently than binary ones; moreover, when a study’s goal is not to rank individuals but to test a theory-driven model (e.g., using scores as an outcome), latent scores are often preferable to sum scores for precision (McNeish, 2024). For example, Schmalbach et al. (2025) showed that, in terms of effect sizes for group mean differences, effect sizes based on observed scores can substantially over- or under-estimate true group differences.

More broadly, for theory validation where the measurement model itself is part of the inferential claim or for high-stakes decisions that require strict precision and minimal error, the information that sum scores cannot provide becomes crucial (e.g., measurement invariance across groups). In these contexts, more complex approaches are not over-specification but often necessary: SEM frameworks, invariance testing, and models that explicitly incorporate error variance (e.g., residual covariance) can directly address threats that sum-score approaches typically leave unmodeled. In short, sum scores are not inherently bad and can be an excellent choice under constraints, but when validity and precision are the core priorities, their simplicity can become a risk.

2.1.4 Statistical Implications of the Debate

For more than a century, the sum score has been used as an easy and practical tool for communication and interpretability, especially for audiences without professional training in measurement. Mislevy (2024) described the intuitive test theory idea of “quantity”. The appeal of the sum score often comes from this human intuition about measuring “more vs. less,” typically grounded in a CTT framework and sometimes supported by other statistical modeling. Beyond interpretability, as highlighted above sum scores can facilitate

replicability, and under certain conditions yield results comparable to latent variable scores in practice (Sijtsma et al., 2024b; Widaman and Revelle, 2024).

However, the use of sum scores should be justified by evidence of adequate measurement quality and validity. Reliability can remain high even when unidimensionality is violated (Cortina, 1993); instead, it reflects how consistently indicators behave, which is not a sufficient condition for validity. As Clark and Watson (2019) noted, reliability is a necessary but not sufficient condition for unidimensionality. Therefore, reliability or the inherent advantages of sum scores should not be the sole argument for their use.

When construct validity is confirmed via scale validation with factor analysis, it should be verified with acceptable goodness-of-fit and moderate to large factor loadings implying similar high weights across items. In contrast, latent variable modeling encourages explicit modeling of the relationship between multiple indicators and latent constructs, testing dimensionality, verifying measurement invariance, and giving more precise estimates all of which support richer validity evidence (McNeish, 2024). Ultimately, whether a particular score choice is justified depends on the purpose of the scores and the assumptions that are defensible in that context.

2.2 Longitudinal Growth Curve Modeling

Latent growth curve modeling (LGCM) is one of the most widely used models for analyzing longitudinal data to capture the growth trajectories (Bishop et al., 2015; Meredith and Tisak, 1990). To this end, the second-order LGCM (2-LGCM; Geiser et al., 2013, also called as curve-of-factors model (McArdle, 1988), can model the growth of latent factors while explicitly accounting for measurement error. Despite these advantages, most applied studies have traditionally fitted first-order LGCMs (1-LGCM) using sum or mean scores (Isiordia and Ferrer, 2018). We will cover the two types of LGCM: 1-LGCM using sum-scores and 2-LGCM by latent variables. For the current study, only linear growth model is considered.

2.2.1 Second-Order LGCM (2-LGCM): Specification and Identification

Unlike the 1-LGCM, which assumes error-free measurement of the observed sum variables, the 2-LGCM accounts for measurement error within the latent construct (Geiser et al., 2013). Suppose we observe 5 items at each time point 5 repeatedly. For the person on item i at time t , let $\mathbf{y}$ denotes 5×5 ($i = 5, t = 5$) response vector for a person, stacked as items within time:

$$\mathbf{y} = \begin{bmatrix} y_{11} \\ \vdots \\ y_{51} \\ y_{12} \\ \vdots \\ y_{55} \end{bmatrix} = \begin{bmatrix} \tau_{11} \\ \vdots \\ \tau_{51} \\ \tau_{12} \\ \vdots \\ \tau_{55} \end{bmatrix} + \begin{bmatrix} \lambda_{11} & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ \lambda_{51} & 0 & \cdots & 0 \\ 0 & \lambda_{12} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_{55} \end{bmatrix} \begin{bmatrix} \eta_1 \\ \eta_2 \\ \vdots \\ \eta_5 \end{bmatrix} + \begin{bmatrix} \varepsilon_{11} \\ \vdots \\ \varepsilon_{51} \\ \varepsilon_{12} \\ \vdots \\ \varepsilon_{55} \end{bmatrix} \quad (4)$$

In matrix form, the measurement model is

$$\mathbf{y} = \boldsymbol{\tau} + \boldsymbol{\Lambda} \boldsymbol{\eta}^{(\text{gen})} + \boldsymbol{\varepsilon}, \quad (5)$$

where $\boldsymbol{\tau}$ is a 25×1 vector of measurement intercepts, $\boldsymbol{\Lambda}$ is a 25×5 block-diagonal factor loading matrix, and $\boldsymbol{\varepsilon}$ is a 25×1 residual vector with $\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$ and $\text{Var}(\boldsymbol{\varepsilon}) = \boldsymbol{\Theta}$.

The conditional mean of $\mathbf{y}$ given the latent factor vector $\boldsymbol{\eta}^{(\text{gen})}$ is

$$\mathbb{E}[\mathbf{y} \mid \boldsymbol{\eta}^{(\text{gen})}] = \boldsymbol{\tau} + \boldsymbol{\Lambda} \boldsymbol{\eta}^{(\text{gen})}, \quad (6)$$

and the corresponding conditional variance is $\text{Var}[\mathbf{y} \mid \boldsymbol{\eta}^{(\text{gen})}] = \boldsymbol{\Theta}$. The marginal (model-implied) variance of $\mathbf{y}$ is

$$\text{Var}(\mathbf{y}) = \boldsymbol{\Lambda} \text{Var}(\boldsymbol{\eta}^{(\text{gen})}) \boldsymbol{\Lambda}' + \boldsymbol{\Theta}. \quad (7)$$

Now, we can write latent constructs in data-generating model $\boldsymbol{\eta}^{(gen)}$ by time as follows

$$\boldsymbol{\eta}^{(gen)} = \begin{bmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \\ \eta_4 \\ \eta_5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} * \begin{bmatrix} \xi_I \\ \xi_S \end{bmatrix} + \begin{bmatrix} \zeta_1 \\ \zeta_2 \\ \zeta_3 \\ \zeta_4 \\ \zeta_5 \end{bmatrix}. \quad (8)$$

In a matrix form,

$$\boldsymbol{\eta}^{(gen)} = \mathbf{B}\boldsymbol{\xi}^{(gen)} + \boldsymbol{\zeta}^{(gen)}, \quad (9)$$

where $\mathbf{B}$ is 5×2 basis matrix $[\mathbf{1}_5, (0, 1, \dots, 5-1)]'$ encoding linear growth by one unit increase, $\boldsymbol{\xi}^{(gen)}$ contains growth intercept and slope factors $[\xi_I, \xi_S]'$, with I and S refer to intercept and slope each, and $\boldsymbol{\zeta}^{(gen)}$ is a vector time-specific disturbance. The vector of latent factor's intercepts is not included here for readability because it is fixed as a zero for direct interpretation of means in growth parameters (Yang et al., 2021).

Then latent factors' mean and slope are

$$\mathbb{E}[\boldsymbol{\eta}^{(gen)}] = \mathbf{B}\boldsymbol{\kappa}^{(gen)}, \quad (10)$$

$$\text{Var}[\boldsymbol{\eta}^{(gen)}] = \mathbf{B}\boldsymbol{\Phi}^{(gen)}\mathbf{B}' + \boldsymbol{\Psi}^{(gen)}, \quad (11)$$

where $\boldsymbol{\kappa}^{(gen)} = [\kappa_I, \kappa_S]'$ is the mean vector of the growth factors, $\boldsymbol{\Phi}^{(gen)}$ is the 2×2 covariance matrix of $\boldsymbol{\xi}^{(gen)}$, and $\boldsymbol{\Psi}^{(gen)}$ is a 5×5 covariance matrix of the disturbances $\boldsymbol{\zeta}^{(gen)}$. The path diagram of 2-LGCM is provided at Figure 1.

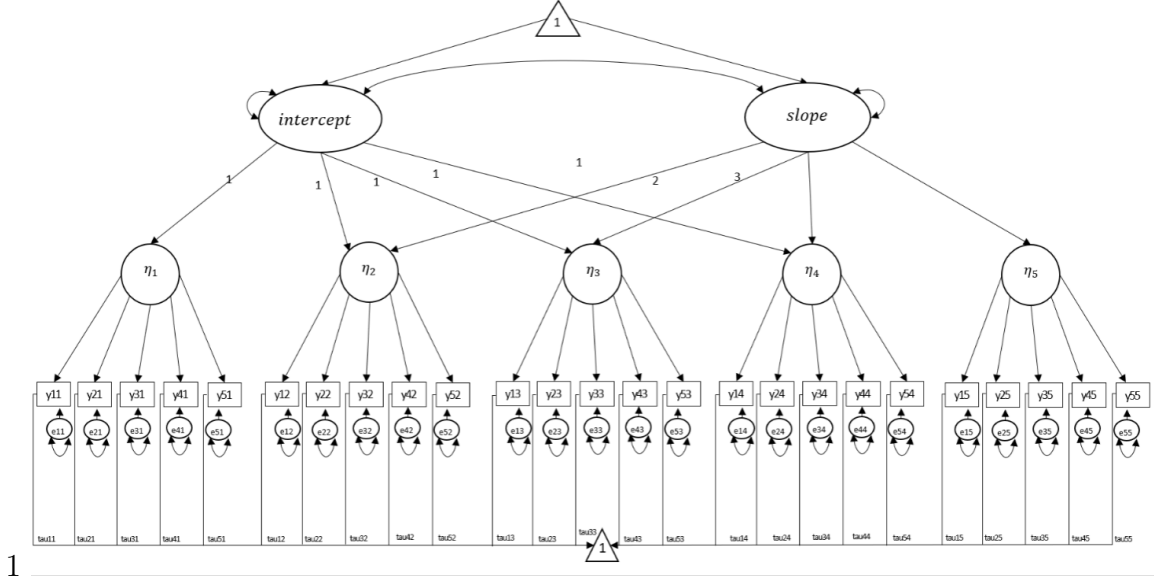


Figure 1. Path diagram of 2-LGCM.

Note. Figure and mathematical expressions are adapted from Grimm et al. (2016) and Yang et al. (2021).

Identification of the 2-LGCM is critical, particularly when it is compared with the sum-score 1-LGCM, because the choice of identification method directly affects scale alignment with the sum-score 1-LGCM in later sections. Three identification methods are used in the 2-LGCM: (1) marker-variable approach, (2) effect-coding approach, and (3) fixed-variance approach. The marker-variable approach fixes one factor loading to 1 and the corresponding intercept to 0 for a designated reference indicator at each time point. The effect-coding approach instead constrains the loadings to constant (usually 1, so the sum of loadings to be 5, the number of indicators) and the intercepts to average 0 (sum to 0) at each wave (Little et al., 2006). The fixed-variance approach sets the latent factor variance to 1 and the latent factor mean to 0 at one time point; the variances and means at the remaining waves are freely estimated and identified through these constraints, so that the second-order growth parameters remain estimable. Among the three, the marker-variable approach places η_t on a specific indicator's metric (e.g., the first item whose loading fixed as 1), which is what makes a direct scale comparison with the raw sum

score aggregating all indicators difficult.

2.2.2 Sum-Score 1-LGCM: Specification and What It Estimates

While the 2-LGCM fits the trajectory to the latent factor measured by indicators at each time point, 1-LGCM directly models growth trajectory at the level of observed scores. Sum scores at time t is created by aggregating each indicators from generating model equation (5) as:

$$S_t = \sum_{i=1}^I y_{it} = \sum_{i=1}^I (\tau_{it} + \lambda_{it} \eta_t^{(\text{gen})} + \varepsilon_{it}). \quad (12)$$

For notational convenience, define

$$c_t = \sum_{i=1}^I \tau_{it}, \quad a_t = \sum_{i=1}^I \lambda_{it}. \quad (13)$$

Then the sum score and its expectation can be written as

$$S_t = c_t + a_t \eta_t^{(\text{gen})} + \sum_{i=1}^I \varepsilon_{it}, \quad (14)$$

$$\text{E}[S_t] = c_t + a_t \text{E}[\eta_t^{(\text{gen})}]. \quad (15)$$

Stacking (14) over the time points yields $\mathbf{S} = [S_1, \dots, S_5]'$, which takes the form of 1-LGCM:

$$\mathbf{S} = \mathbf{B} \boldsymbol{\xi} + \mathbf{e}, \quad (16)$$

where $\mathbf{B}$ is the 5×2 linear basis matrix identical to (9), $\boldsymbol{\xi} = [\xi_I, \xi_S]'$ contains the growth intercept and slope factors, and $\mathbf{e} = [e_1, \dots, e_5]'$ is the residual vector with $\text{E}[\mathbf{e}] = \mathbf{0}$ and $\text{Var}[\mathbf{e}] = \boldsymbol{\Theta}_S$. Note that the observed-variable intercepts are fixed to zero in standard 1-LGCM identification; the time-varying composite intercept c_t and scale a_t are absorbed into the estimated growth-factor means $\boldsymbol{\kappa}$. The growth factor mean and variance-covariance structure are expressed as $\boldsymbol{\kappa} = \text{E}[\boldsymbol{\xi}] = [\boldsymbol{\kappa}_1, \boldsymbol{\kappa}_2]'$, and $\boldsymbol{\Phi} = \text{Var}[\boldsymbol{\xi}]$. A

critical assumption underlying this model is that the 1-LGCM treats each sum score S_t as a construct invariant in meaning over time. If item intercepts or loadings shift over time (i.e., measurement non-invariance), c_t and a_t become time-varying, and observed change in S_t no longer reflects latent growth alone. To see exactly how these measurement shifts affect the observed change, consider the difference in expected sum scores between adjacent time points from (15):

$$\begin{aligned} E[S_{t+1}] - E[S_t] &= (c_{t+1} + a_{t+1} E[\eta_{t+1}^{(gen)}]) - (c_t + a_t E[\eta_t^{(gen)}]) \\ &= a_{t+1}(E[\eta_{t+1}^{(gen)}] - E[\eta_t^{(gen)}]) + (a_{t+1} - a_t) E[\eta_t^{(gen)}] + (c_{t+1} - c_t). \end{aligned} \tag{17}$$

The first term captures the change in the latent variable scaled by a_{t+1} . The second term reflects the drift in the sum of factor loadings, weighted by the latent level at time t . The third term reflects the drift in the sum of item intercepts. Consequently, violations of longitudinal measurement invariance become entangled with the observed mean change. Even when the true latent change is zero ($E[\eta_{t+1}] - E[\eta_t] = 0$), a non-zero loading drift or intercept drift will produce spurious growth in the sum score. This confounding vanishes only if non-invariance happens to cancel across items out such that a_t and c_t remain constant over time, which is unlikely to hold in practice often.

2.3 Measurement Invariance in Longitudinal Growth Models

Examining measurement invariance in LGCM is critical because the estimated growth trajectory should reflect genuine change in the latent construct, not artifacts arising from shifts in measurement properties over time (Kim et al., 2017). In practice, however, invariance testing is seldom conducted due to limited sample sizes and the relatively complex modeling procedure compared to fitting a single model. Nevertheless, violation of measurement invariance across time implies that the operational definition of the latent construct has fundamentally changed. For example, Fried et al. (2016) found that a depression scale administered to a large sample ($n = 3,509$) satisfied neither

unidimensionality nor measurement invariance, warning against the use of sum scores derived from a non-invariant scale. To ensure that the latent variable being measured retains a consistent meaning over time without change in its internal structure, it is essential to verify whether longitudinal measurement invariance holds, either fully or partially.

Measurement invariance has been frequently explored as a tool for multi-group analysis examining factorial invariance in factor analysis, similarly to differential item functioning in IRT (Liu et al., 2017; Tay et al., 2015). Typically, a sequence of four measurement invariance tests is conducted (Meredith, 1993; Putnick and Bornstein, 2016). Configural invariance evaluates whether the same factor structure holds across groups or time points by fitting the same indicators measured by the same latent factor to both occasions or groups. Metric (weak) invariance tests whether factor loadings are equal across occasions. Using the factor analysis notation from equation (3), $\lambda_{ig} = \lambda_{ik}$, $g \neq k$ where g and k refer to groups. Scalar (strong) invariance further constrains intercepts to be equal, $\tau_{ig} = \tau_{ik}$, $g \neq k$. Strict invariance additionally restricts residual variances to be the same, $\theta_{ig} = \theta_{ik}$, $g \neq k$. This framework can be extended to the longitudinal setting to examine measurement invariance across multiple time points, instead of multiple groups (Widaman et al., 2010).

An evaluation of measurement invariance proceeds by comparing model fit across successively constrained models. As standard practice, a likelihood ratio test (LRT) is conducted by evaluating the chi-square difference ($\Delta\chi^2$) between a baseline model and a more restrictive nested model. Because the LRT is sensitive to large sample sizes, differences in approximate fit indices (AFIs) have been suggested as complementary criteria (Chen, 2007; Cheung & Rensvold, 2002). The change in AFIs between models is evaluated against a priori cutoffs to determine whether the additional constraints produce a practically meaningful degradation in fit (e.g., evaluating MNI in factor loadings using $\Delta CFI \leq .010$ and $\Delta RMSEA \leq .015$; Chen, 2007), although the generalizability of these

cutoffs across different modeling conditions remains limited.

2.4 Correlated Uniqueness in Longitudinal Models

When correlated uniquenesses are specified in the model, the residuals of the same item are correlated across different time points, such that $\text{Cov}(\boldsymbol{\varepsilon}_t, \boldsymbol{\varepsilon}_k) \neq \mathbf{0}$ for $t \neq k$. Although ignoring existing correlated uniqueness is known to yield biased parameter estimates (Murphy et al., 2011) and the presence of carry-over effects is often a reasonable assumption (Grimm & Widaman, 2010), these correlations are frequently omitted from models to maintain parsimony. Schmalbach et al. (2025) also emphasized that the impact of unmodeled residual covariances in SEM should be scrutinized, particularly when compared to results derived from observed sum scores.

Correlated uniqueness poses an additional and independent source of contamination to sum-score growth models even when measurement invariance holds perfectly. With correlated uniqueness, the sum-score 1-LGCM cannot accommodate the cross-time residual covariance, which instead will be absorbed into the growth parameter estimates as bias. We denote this cross-time residual covariance matrix as $\boldsymbol{\Theta}_{tk}$. Because different items are assumed to have uncorrelated residuals, $\boldsymbol{\Theta}_{tk}$ is diagonal matrix containing same item residuals covary across time. Following the (14), covariance structure between two sum scores at time t and k is $\text{Cov}(S_t, S_k) = \text{Cov}(a_t \eta_t^{(\text{gen})} + \sum \varepsilon_{it}, a_k \eta_k^{(\text{gen})} + \sum \varepsilon_{ik})$. Under this notation, the population covariance structure of sum scores becomes

$$\text{Cov}(S_t, S_k) = a_t a_k \text{Cov}(\eta_t^{(\text{gen})}, \eta_k^{(\text{gen})}) + \sum_{i=1}^5 \text{Cov}(\varepsilon_{it}, \varepsilon_{ik}). \quad (18)$$

When $t \neq k$, the second term equals $\text{tr}(\boldsymbol{\Theta}_{tk})$, the sum of all item-level correlated uniquenesses across the two time points (t, k) . Because $\text{tr}(\boldsymbol{\Theta}_{tk})$ enters additively and does not involve a_t or a_k , it contributes to the off-diagonal covariance of sum scores independently of the latent factor covariance.

Now, the model-implied covariance between sum scores is provided following (16) as:

$$\begin{aligned}\widehat{\text{Cov}}(S_t, S_k) &= \text{Cov}(\xi_I + (t-1)\xi_S, \xi_I + (k-1)\xi_S) \\ &= \hat{\Phi}_{II} + (t+k-2)\hat{\Phi}_{IS} + (t-1)(k-1)\hat{\Phi}_{SS}, \quad t \neq k.\end{aligned}\tag{19}$$

Note that the population covariance between latent factors is also

$$\text{Cov}(\eta_t^{(\text{gen})}, \eta_k^{(\text{gen})}) = \Phi_{II}^{(\text{gen})} + (t+k-2)\Phi_{IS}^{(\text{gen})} + (t-1)(k-1)\Phi_{SS}^{(\text{gen})}, \quad t \neq k,$$

which has exactly the same form as Equation (19). This means that the sum-score 1-LGCM treats S_t as if it were latent variable η_t itself. Comparing the population covariance in equation (18) with the model-implied form in equation (19), the 1-LGCM omits two components that are present in the true covariance structure: the scaling factor $a_t a_k$ that links the sum-score covariance to the latent covariance, and the correlated uniqueness term $\text{tr}(\Theta_{tk})$. Because the model has no structure to represent either component, both are absorbed into $\hat{\Phi}$ as bias. When correlated uniqueness arises, the sum score is no longer a stable linear composite of the latent factor, and the observed mean trajectory can diverge from the true latent trajectory. This bias is inherent to the sum-versus-latent distinction and does not require any drift in measurement properties.

Chapter 3 Metric Equivalence Between 1-LGCM and 2-LGCM

Before presenting the research questions and simulation studies, one preliminary issue must be addressed: the parameter estimates from the sum-score 1-LGCM and the latent-variable 2-LGCM are expressed on different metrics, and any direct comparison between them would confound genuine model differences with metric artifacts. The 1-LGCM operates on observed sum scores, whose metric is jointly determined by the number of items and their measurement parameters, whereas the 2-LGCM estimates growth parameters on the latent factor’s own scale, which depends on how the model is identified. Unless these two sets of estimates are placed onto a common metric, it is impossible to determine whether a discrepancy reflects a real difference in how the models recover growth or merely a difference in units. This chapter derives the metric-alignment transformation that places the 2-LGCM estimates onto the sum-score metric via effect-coding identification. Under strict measurement invariance, this transformation establishes an exact affine equivalence between the two models, providing the analytic foundation on which the simulation conditions in subsequent Study 1 are built. The analytical relationship of how measurement non-invariance disrupts this equivalence is deferred to Appendix A; the following simulation Study 2 evaluate its practical consequences empirically.

3.1 Transformation to Effect-Coding Fitted Population

To align the parameters from 1-LGCM and 2-LGCM on the same scale, effect-coding (EC) is used and its population parameters are transformed from the population-generating model. This transformation is motivated by Yang et al. (2021)’s work on effect-coding and Deng and Yuan (2023)’s remarks on rescaling. EC identifies the scale of the latent variable such that, for each measurement time point, the factor loadings and intercepts satisfy

$$\sum_{i=1}^I \lambda_{it}^{(\text{ec})} = 1, \quad \sum_{i=1}^I \tau_{it}^{(\text{ec})} = 0, \quad (20)$$

where (ec) denotes the parameters are in the effect-coding fitted metric.

Here, assume there is a linear transformation between the two latent factor scales from the generating model to the effect-coding model as:

$$\eta_t^{(\text{ec})} = w_t \eta_t^{(\text{gen})} + \delta_t, \quad (21)$$

There are two unknown parameters: w_t is a weight and δ_t is a location parameter. With this relationship, regardless of how the generating model's metric is specified, this transformation maps the latent factors onto the EC metric, making the parameter estimates from both models directly comparable. From equation (21), $\eta_t^{(\text{gen})} = \frac{\eta_t^{(\text{ec})} - \delta_t}{w_t}$, and now equation (6) can also be re-expressed in the EC-fitted metric as:

$$\mathbb{E}[\mathbf{y}_t \mid \eta_t^{(\text{gen})}] = \boldsymbol{\tau}_t + \boldsymbol{\Lambda}_t \eta_t^{(\text{gen})} = \boldsymbol{\tau}_t + \boldsymbol{\Lambda}_t \frac{\eta_t^{(\text{ec})} - \delta_t}{w_t} = \underbrace{\boldsymbol{\tau}_t - \frac{\delta_t}{w_t} \boldsymbol{\Lambda}_t}_{\boldsymbol{\tau}_t^{(\text{ec})}} + \underbrace{\frac{\boldsymbol{\Lambda}_t}{w_t}}_{\boldsymbol{\Lambda}_t^{(\text{ec})}} \eta_t^{(\text{ec})}.$$

Then,

$$\boldsymbol{\Lambda}_t^{(\text{ec})} = \frac{\boldsymbol{\Lambda}_t}{w_t}, \quad \boldsymbol{\tau}_t^{(\text{ec})} = \boldsymbol{\tau}_t - \frac{\delta_t}{w_t} \boldsymbol{\Lambda}_t. \quad (22)$$

Here, imposing the effect-coding constraints determines w_t and δ_t , by substituting equation (22) into the constraints in equation (20) and solving for w_t and δ_t :

$$w_t = \sum_{i=1}^I \lambda_{it}, \quad \delta_t = \sum_{i=1}^I \tau_{it}. \quad (23)$$

These are precisely a_t and c_t defined in equation (13); henceforth we use a_t and c_t .

Substituting into equation (21):

$$\eta_t^{(\text{ec})} = \sum_{i=1}^I \lambda_{it} \eta_t^{(\text{gen})} + \sum_{i=1}^I \tau_{it} = a_t \eta_t^{(\text{gen})} + c_t. \quad (24)$$

The effect-coded latent factor is now achieved by linear transformation of the factors from

the generating model.

3.2 Connection to Sum-Score 1-LGCM

Using the measurement part from the generating model (14), the conditional mean of sum scores given the latent factor is written as

$$E[S_t | \eta_t^{(\text{gen})}] = \mathbf{1}'_I \boldsymbol{\tau}_t + \mathbf{1}'_I \boldsymbol{\Lambda}_t \eta_t^{(\text{gen})} = c_t + a_t \eta_t^{(\text{gen})} = \eta_t^{(\text{ec})}, \quad (25)$$

where multiplying by $\mathbf{1}'_I$ performs the summation across the vector and $\eta_t^{(\text{ec})}$ is defined in equation (24). Then, the sum score can be decomposed as

$$S_t = \eta_t^{(\text{ec})} + \varepsilon_t^{(\text{sum})}. \quad (26)$$

In a nutshell, sum-score 1-LGCM recovers the growth parameters of the EC-fitted latent scale $\eta_t^{(\text{ec})}$ rather than those of the original generating factor $\eta_t^{(\text{gen})}$. That is how effect-coding identification strategy puts the 2-LGCM and the sum-score 1-LGCM on the same underlying scale.

3.3 Implicit Metric Choice in Sum-Score

Sum scores appears to be a metric-neutral decision because it doesn't seem to specify any measurement model explicitly. For example, the sum-score 1-LGCM simply fixes unit loadings on the intercept factor and assigns equally spaced coefficients $(0, 1, \dots, T - 1)$ to the slope factor, without requiring any specific scaling or identification strategy. However, this impression is misleading because parameter estimates from the sum-score 1-LGCM are implicitly expressed in the effect-coded metric. As equation (26) demonstrated, both simple aggregation and explicit effect-coding constraints arrive at the same metric, despite originating from fundamentally different procedures.

This equivalence in the latent factor and observed sum-score, however, can be extended

to the growth factor parameters with a important condition: measurement invariance. When loadings and intercepts are constant across time, the linear transformation links the generating-model growth factors to the effect-coded growth factors. The sum-score is then a noisy realization of the same latent construct, and the 1-LGCM and 2-LGCM recover the same EC-fitted growth parameters up to sampling error. Once measurement parameters drift across time, this alignment breaks. The nature and severity of this breakdown depend on which parameter is non-invariant and how many items are affected, the questions that simulation Study 2 in the following chapter is designed to answer (see Appendix A for the formal derivation).

Chapter 4 Research Questions

The sum-score 1-LGCM and the latent-variable 2-LGCM operate on different metrics, even though both use the same indicators. This is because the two models handle measurement residuals differently, and the act of summation transforms the implicit metric of the composite into one that matches the effect-coded 2-LGCM only under strict measurement invariance (MI). Both models produce essentially the same growth parameter estimates when (1) the measurement model is correctly specified and (2) all measurement parameters remain constant across time. The current dissertation examines what happens when each premise is violated separately, through two simulation studies.

The first study retains the second premise by assuming strict MI. Instead, it examines the first premise: correct specification of the measurement model. When residual covariances exist between the same indicator across time points (e.g., $\text{cor}(y_{i1t1}, y_{i1t2}) \neq 0$), the sum-score 1-LGCM cannot model this residual component systematically because it uses aggregated scores. By its nature, the sum-score 1-LGCM is inherently misspecified, as it fails to account for these correlated residuals. Thus, Study 1 investigates how measurement model misspecification caused by ignoring correlated uniquenesses may produce differential bias in growth parameter estimates.

The second premise, that measurement parameters remain invariant over time, can also be violated in applied settings. When measurement non-invariance (MNI) is present in the factor loadings or intercepts, sum scores may represent different latent constructs across time points, both theoretically and mathematically. Because this equivalence can be broken under MNI, the scale alignment of the 1-LGCM and the 2-LGCM no longer holds. Accordingly, the second study investigates how growth average parameter estimates differ between the 1-LGCM and the 2-LGCM under MNI, and whether scale-free metrics can capture the diverging results across different types of MNI. All research questions are addressed through Monte Carlo simulation studies, which allow for systematic control over the data-generating conditions.

Chapter 5 Correlated Uniqueness as a Potential Confound: Study 1

5.1 Purpose and Research Questions

The first study investigates the impact of correlated uniqueness on the recovery of latent growth curve model parameters, specifically examining the effect of including versus excluding the measurement error component. To achieve this, we compare the performance of (1) a first-order LGCM (1-LGCM) using sum scores and (2) a second-order LGCM (2-LGCM) using latent variables. This comparison is conducted by manipulating the presence of correlated uniqueness in the data-generating process, all while maintaining a condition of strict measurement invariance. Also, to align the scales under measurement invariance, we will use the effect-coding fitted 2-LGCM. The research questions for Study 1 include:

- RQ1.1: Under strict measurement invariance, how does ignoring correlated uniqueness in the sum-score 1-LGCM affect the recovery of growth parameter estimates relative to the 2-LGCM, and how do sample size and measurement quality moderate this effect?
- RQ1.2: Under the same conditions, how does this manipulation affect the difference in admissibility rates between the two models?

5.2 Method

5.2.1 Data-Generating Model

This section describes the study design and data-generating procedures for the Monte Carlo simulation. The 2-LGCM was used as the data-generation model to define the growth trajectory of the latent construct at the population level. The simulated data were then analyzed using both the 1-LGCM with sum scores and the 2-LGCM with latent variables, to compare the precision and accuracy of growth parameter estimates.

As a running example throughout this dissertation, we use longitudinal data consisting of 5 observed items measured repeatedly across 5 time points. For visual overview of the

data-generating model, see the path diagram in Figure 1. The growth trajectory is modeled as linear, with the first occasion serving as the baseline, such that the slope factor loadings increase gradually by the increase of 1, starting from 0 (i.e., 0, 1, 2, 3, 4).

Population parameter from original data-generating model and effect-coding fitted parameter were presented at Table 1. To ensure the latent scales of both models are

Table 1: Population Parameters for Simulation Study

		Population Parameter Values	
		Original values by data-generating model	Re-scaled values by effect-coding
Growth Parameters			
Mean of growth intercept and slope: $\mathbf{k}' = [k_I, k_S]$		[1, 0.125]	[2.5, 0.25]
Variance/covariance matrix of growth intercept and slope:		Size of ϕ_S^2	
$\phi = \begin{bmatrix} \phi_I^2 & \phi_{IS} \\ \phi_{IS} & \phi_S^2 \end{bmatrix}$	Small	$\begin{bmatrix} 1 & .13 \\ .13 & .10 \end{bmatrix}$	$\begin{bmatrix} 4 & .52 \\ .52 & .41 \end{bmatrix}$
	Medium	$\begin{bmatrix} 1 & .18 \\ .18 & .20 \end{bmatrix}$	$\begin{bmatrix} 4 & .72 \\ .72 & .81 \end{bmatrix}$
	Large	$\begin{bmatrix} 1 & .22 \\ .22 & .30 \end{bmatrix}$	$\begin{bmatrix} 4 & .88 \\ .88 & 1.21 \end{bmatrix}$

aligned on the same metric, I used effect-coding identification for the latent variable models. Both 1-LGCM and 2-LGCM are based on five fixed time points, with five indicators measured repeatedly at each wave. To facilitate the comparison between models, I will evaluate simulation performance using metrics such as statistical power and relative bias. Finally, this study assumes complete data without any missingness.

Regarding the fixed population parameter settings, the means of the growth intercept and slope were fixed at 1 and .125, respectively, to examine the statistical power for detecting a non-zero intercept and slope. The disturbance variance of latent constructs was fixed at $[\psi_1^2, \psi_2^2, \psi_3^2, \psi_4^2, \psi_5^2] = [1.097, .98, .95, .93, .91]$, such that more than 60% of proportion of variance in latent construct was explained by the growth parameters, under each time point. Also, the intercept of indicators is fixed at $\tau_{it} = .1$ and to assume

essentially tau-equivalence, all items factor loadings are the same within and between time points at .4.

5.2.2 Simulation Conditions

Correlated Uniqueness (ρ) The correlated uniqueness (CU) structure for each indicator across time was modeled as a first-order autoregressive pattern with adjacent-wave correlation ρ . Two levels were used: $\rho = 0$ (no CU) and $\rho = 0.15$ (small to moderate size of CU). For the 2-LGCM, adjacent-wave residual covariances for each item were freely estimated when $\rho > 0$; for the 1-LGCM, adjacent-wave covariances of the composite score were likewise freely estimated.

Size of Residual Variance (σ_ϵ^2) To manipulate the levels of measurement quality, residual variance was manipulated by 1.57, .78, and .36, which yielded the poor, good, and very good measurement quality by varying R^2 (.16, .44, .78) accordingly.

Size of Growth Slope Variance (ϕ_{slope}^2) Following previous literature that the reasonable ratio of growth intercept variance to slope variance in practice would be 5:1 (Kim et al., 2017), the data-generating population values were assigned. To change the level of growth slope variance (ϕ_S^2) into small (.102), medium (.202), and large (.303), the growth intercept variance (ϕ_1^2) was fixed at 1 while keeping the ration between 3 to 9. The covariance between the intercept and slope (ϕ_{12}) were set such that the correlation between them would be around .4 which can be considered as small to moderate relationship.

Sample Size (N) Sample sizes varied by 200, 500, and 800 to represent small-to-large conditions commonly considered in Monte Carlo studies of latent growth models.

In total, $108 = 3 (\sigma_\epsilon^2 \text{ size}) \times 3 (\phi_S^2 \text{ size}) \times 2 (\text{correlated uniqueness}) \times 3 (\text{sample size}) \times 2 (\text{Models})$ simulation conditions existed for data generation. Per each condition, 1,000 replications were generated.

5.2.3 Evaluation Measures

First, statistical power was defined as the rejection rate for the growth parameters. For each condition, power was computed as the proportion of replications that rejected the null hypothesis, divided by the total number of replications that produced admissible solutions. To evaluate parameter estimation accuracy, relative bias (RB) was computed as

$$\text{RB} = \frac{\sum_{r=1}^{n_{\text{rep}}} (\hat{\theta}_r - \theta) / \theta}{n_{\text{rep}}},$$

where n_{rep} is the number of replications. RB represents the average bias across replications, expressed as the relative difference between the estimate $\hat{\theta}_r$ and the population value θ . Estimates with RB within $[-.10, .10]$ were considered relatively accurate Muthén and Muthén, 2002. Along with RB, relative root mean square error (RMSE) was also reported. Relative RMSE summarizes average error reflecting both precision and bias, scaled by a reference magnitude, the most commonly absolute true value of the parameter:

$$\text{R.RMSE} = \frac{\sqrt{\frac{1}{n_{\text{rep}}} \sum_{r=1}^{n_{\text{rep}}} (\hat{\theta}_r - \theta)^2}}{|\theta|}. \quad (27)$$

Additionally, 95% coverage rates were calculated as the proportion of replications in which the nominal 95% confidence interval contained the true parameter value. Finally, I compared the proportion of admissible solutions.

5.3 Results

As shown in Figures 2, 3, and 4, the accuracy patterns for the variance and covariance of the growth parameters were consistent across outcomes. In the Basic condition, where correlated uniqueness was absent in both data generation and model estimation, there was essentially no difference between 1-LGCM and 2-LGCM, and the results almost completely overlapped. In contrast, when correlated uniqueness was present (see the Cor.Unique

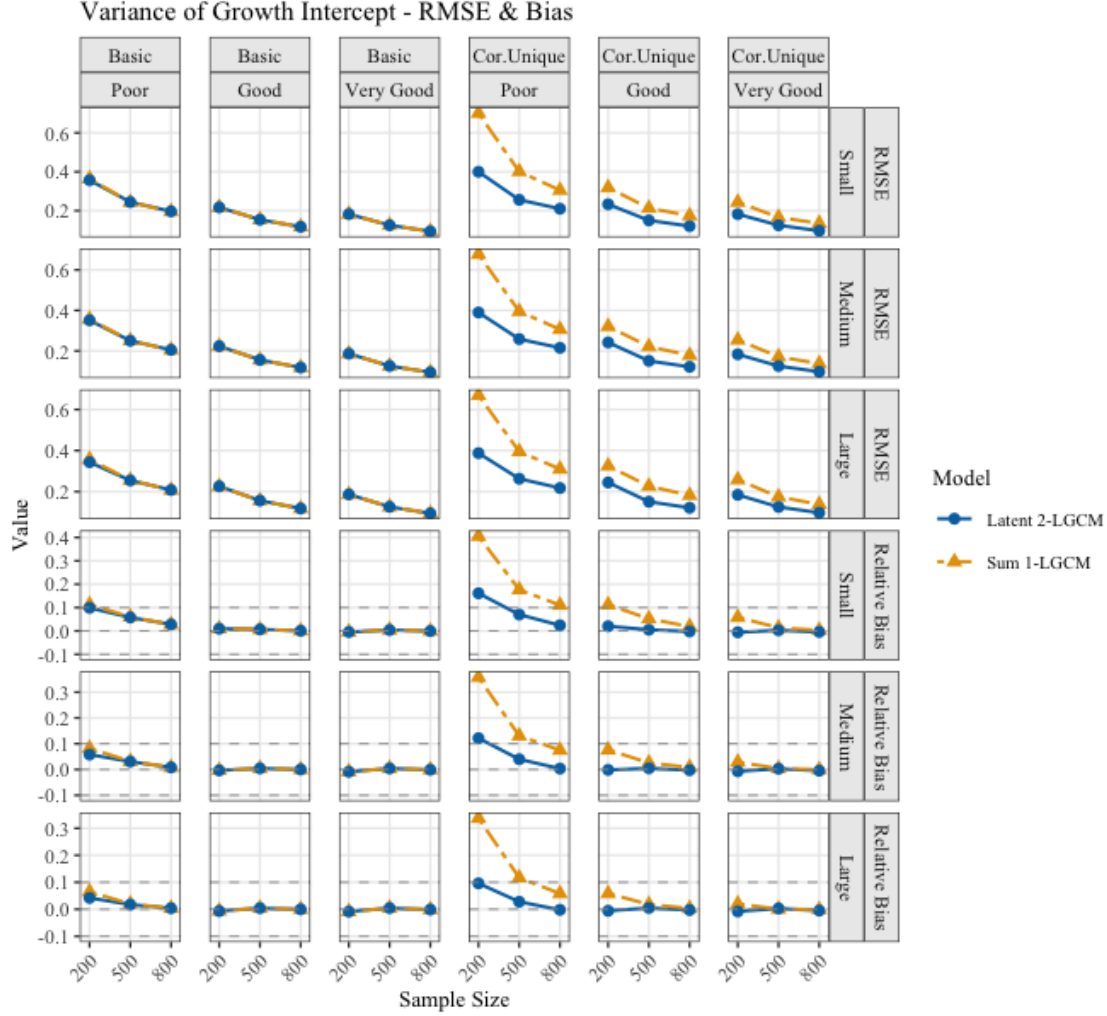


Figure 2. Relative Bias and Relative RMSE on Growth Intercept Variance

Note. Basic = data and model without correlated uniqueness (residual covariance); Cor.Unique = data and model with correlated uniqueness. Poor/Good/Very Good reflect measurement quality (varying via residual variances). Small/Medium/Large denote growth slope variance magnitude respectively.

columns), a clear discrepancy emerged: the latent variable 2-LGCM produced more accurate estimates than the sum score 1-LGCM. Specifically, under correlated uniqueness in 1-LGCM, the variances of the growth intercept (Figure 2) and slope (Figure 3) tended to be slightly over-estimated, whereas the covariance between the intercept and slope was underestimated (Figure 4). This pattern was most pronounced under poor measurement quality, where residual variance was largest and the average R^2 was about .16. As expected, estimation accuracy improved with increasing sample size across all conditions.

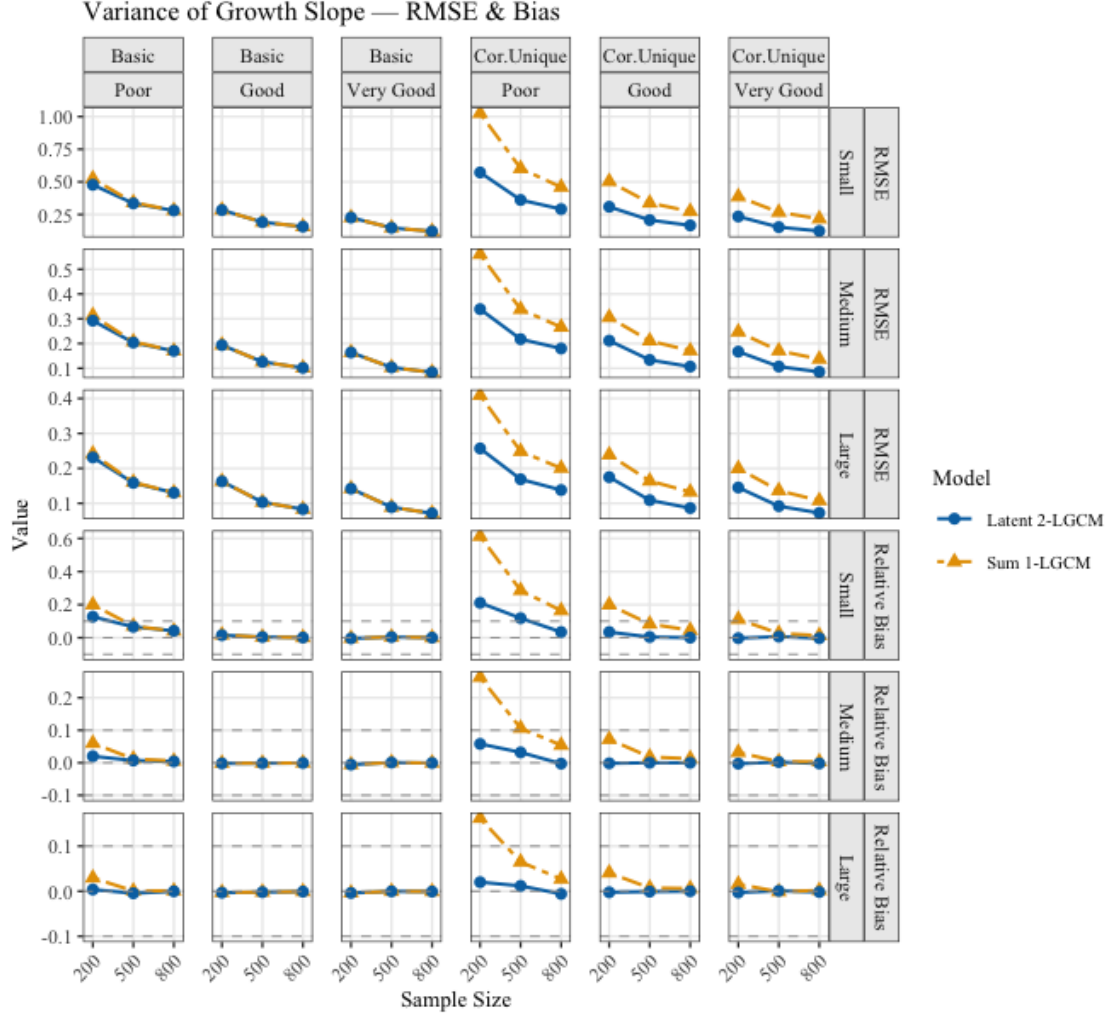


Figure 3. Relative Bias and Relative RMSE on Growth Slope Variance

Note. Basic = data and model without correlated uniqueness (residual covariance); Cor.Unique = data and model with correlated uniqueness. Poor/Good/Very Good reflect measurement quality (varying via residual variances). Small/Medium/Large denote growth slope variance magnitude respectively.

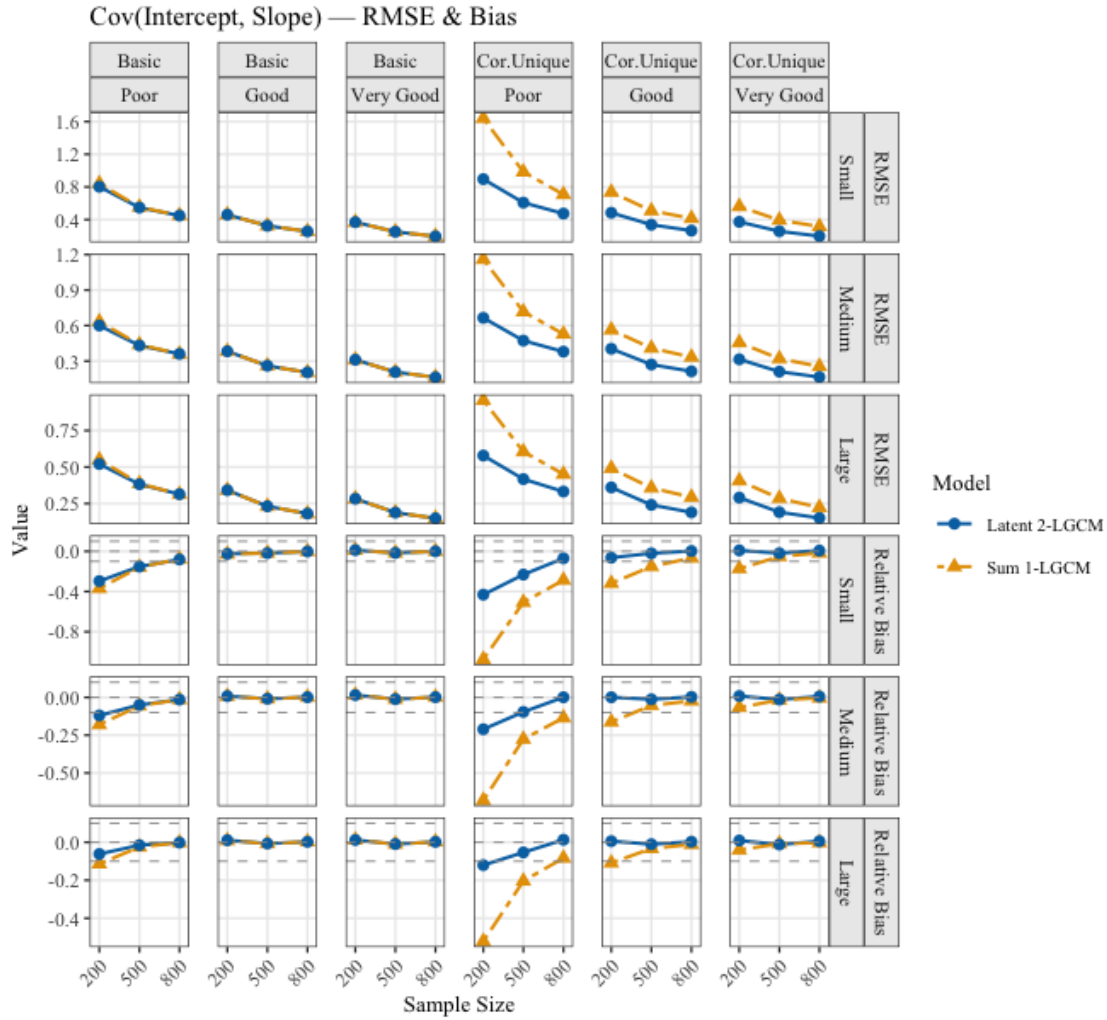


Figure 4. Relative Bias and Relative RMSE on Growth Slope and Intercept Covariance

Note. Basic = data and model without correlated uniqueness (residual covariance); Cor.Unique = data and model with correlated uniqueness. Poor/Good/Very Good reflect measurement quality (varying via residual variances). Small/Medium/Large denote growth slope variance magnitude respectively.

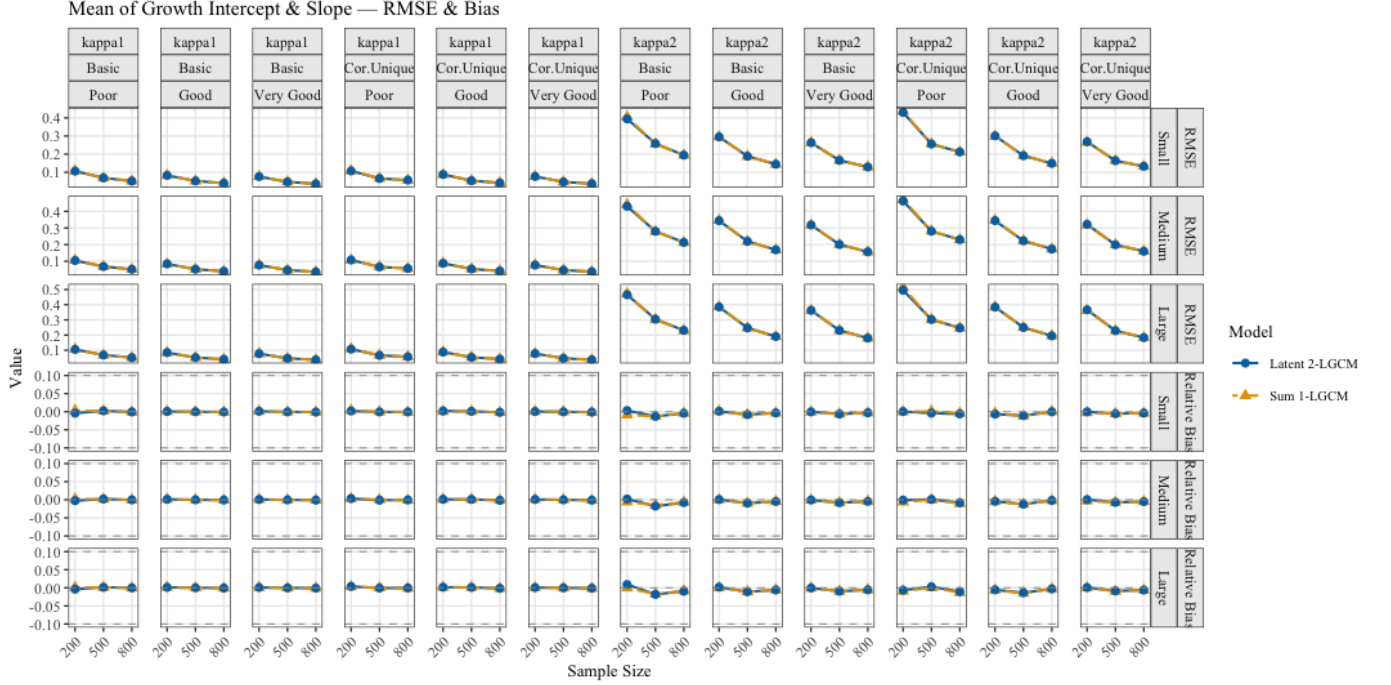


Figure 5. Relative Bias and Relative RMSE on Means of Growth Intercept and Slope

Note. Basic = data and model without correlated uniqueness (residual covariance); Cor.Unique = data and model with correlated uniqueness. Poor/Good/Very Good reflect measurement quality (varying via residual variances). Small/Medium/Large denote growth slope variance magnitude respectively.

The magnitude of the growth slope variance did not produce an obvious difference in accuracy across the simulation conditions. Regarding the means of the growth intercept and slope (Figure 5), there was no meaningful difference in estimation accuracy between the two models, although poor measurement quality still yielded the largest relative RMSE.

For statistical power and 95% coverage rates, the variance and covariance parameters again showed highly similar patterns across Figures 6, 7, and 8. The most substantial differences were driven by the presence and modeling of correlated residuals (Cor.Unique). When correlated residuals were present, power for the growth variances and covariance was higher in 2-LGCM than in 1-LGCM; without correlated residuals, the two models were not distinguishable in power. Notably, several 1-LGCM conditions using sum scores fell below the nominal .80 power threshold, especially under poor measurement quality, even when factor loadings were held constant and the sample size was 500. For 95% coverage, all

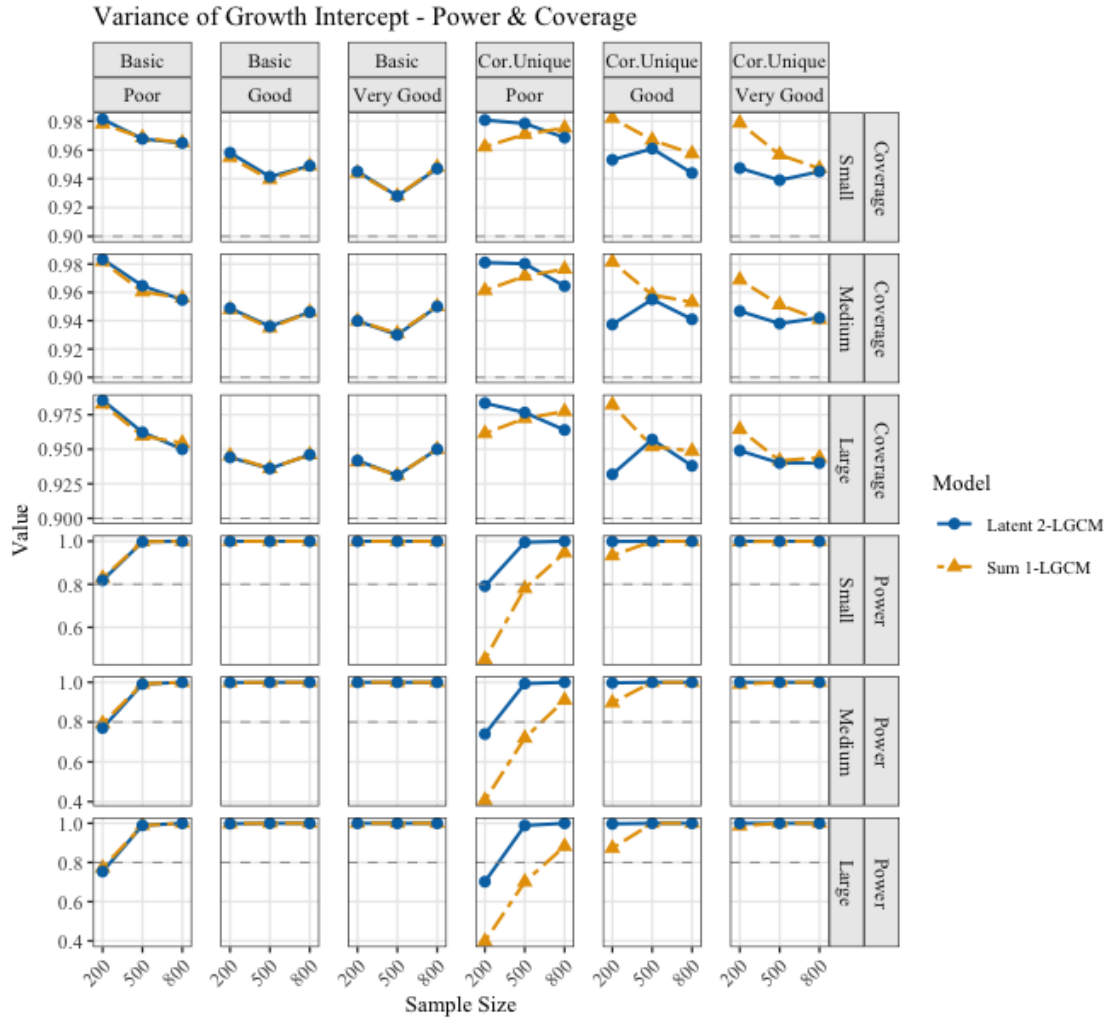


Figure 6. Power and 95% Coverage Rates on Growth Intercept Variance

Note. Basic = data and model without correlated uniqueness (residual covariance); Cor.Unique = data and model with correlated uniqueness. Poor/Good/Very Good reflect measurement quality (varying via residual variances). Small/Medium/Large denote growth slope variance magnitude respectively.

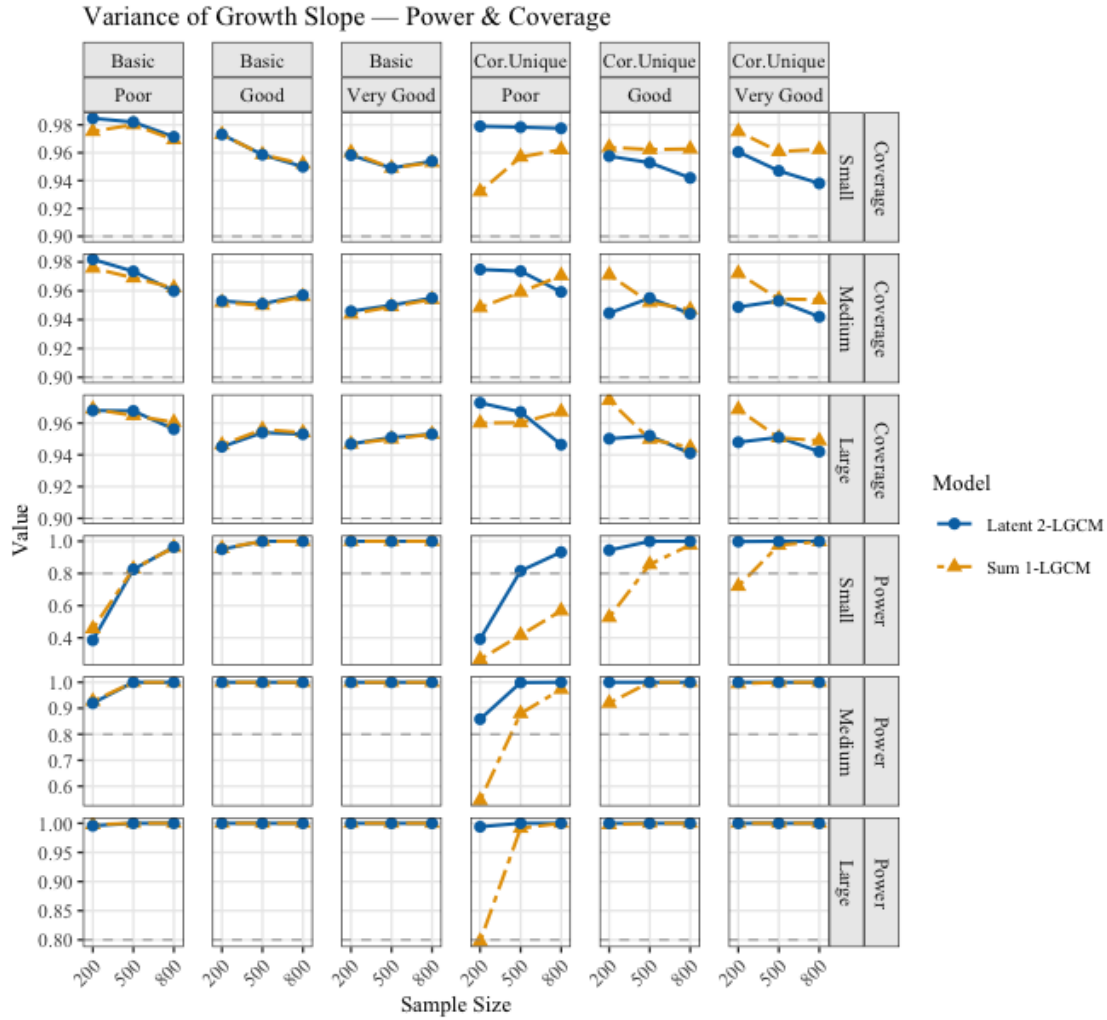


Figure 7. Power and 95% Coverage Rates on Growth Slope Variance

Note. Basic = data and model without correlated uniqueness (residual covariance); Cor.Unique = data and model with correlated uniqueness. Poor/Good/Very Good reflect measurement quality (varying via residual variances). Small/Medium/Large denote growth slope variance magnitude respectively.

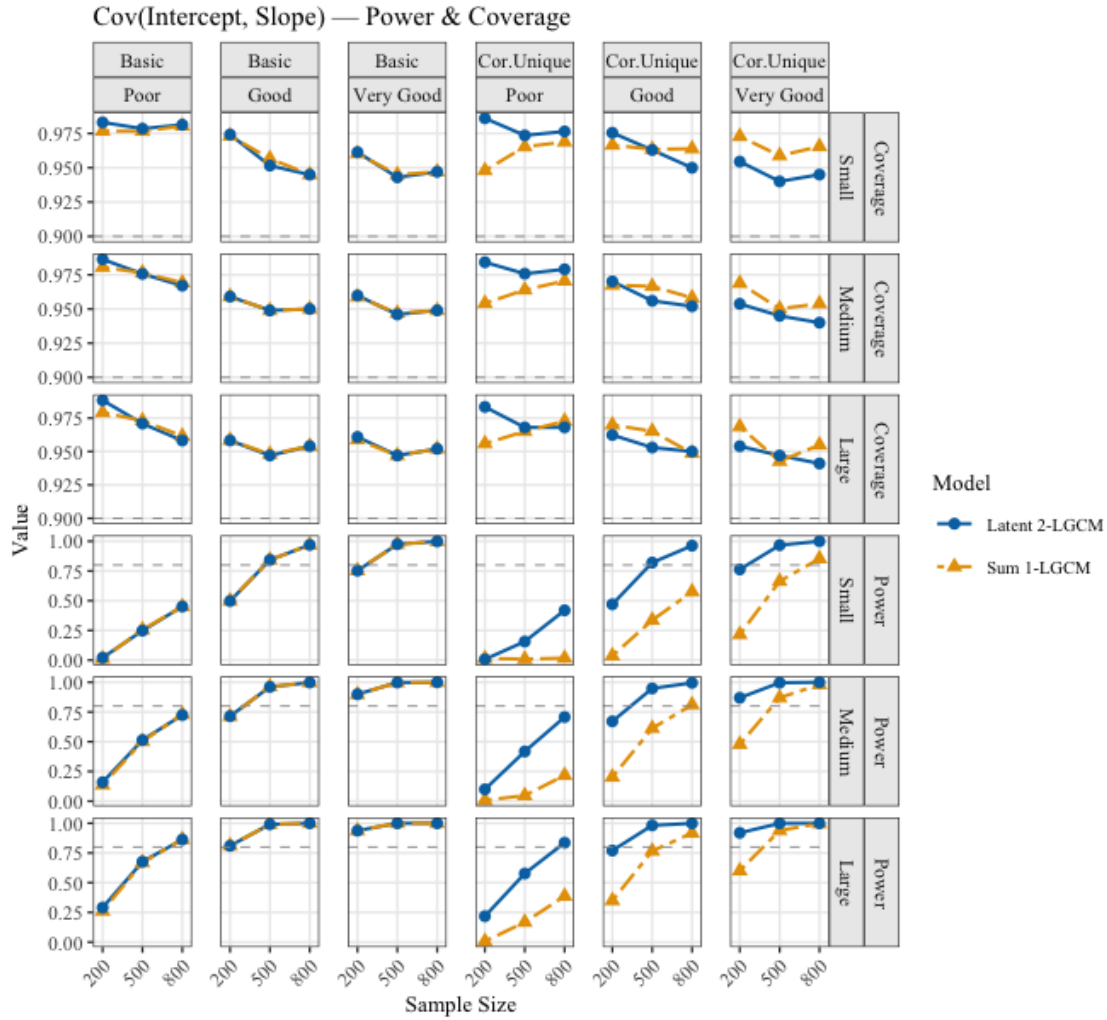


Figure 8. Power and 95% Coverage Rates on Growth Slope Covariance

Note. Basic = data and model without correlated uniqueness (residual covariance); Cor.Unique = data and model with correlated uniqueness. Poor/Good/Very Good reflect measurement quality (varying via residual variances). Small/Medium/Large denote growth slope variance magnitude respectively.

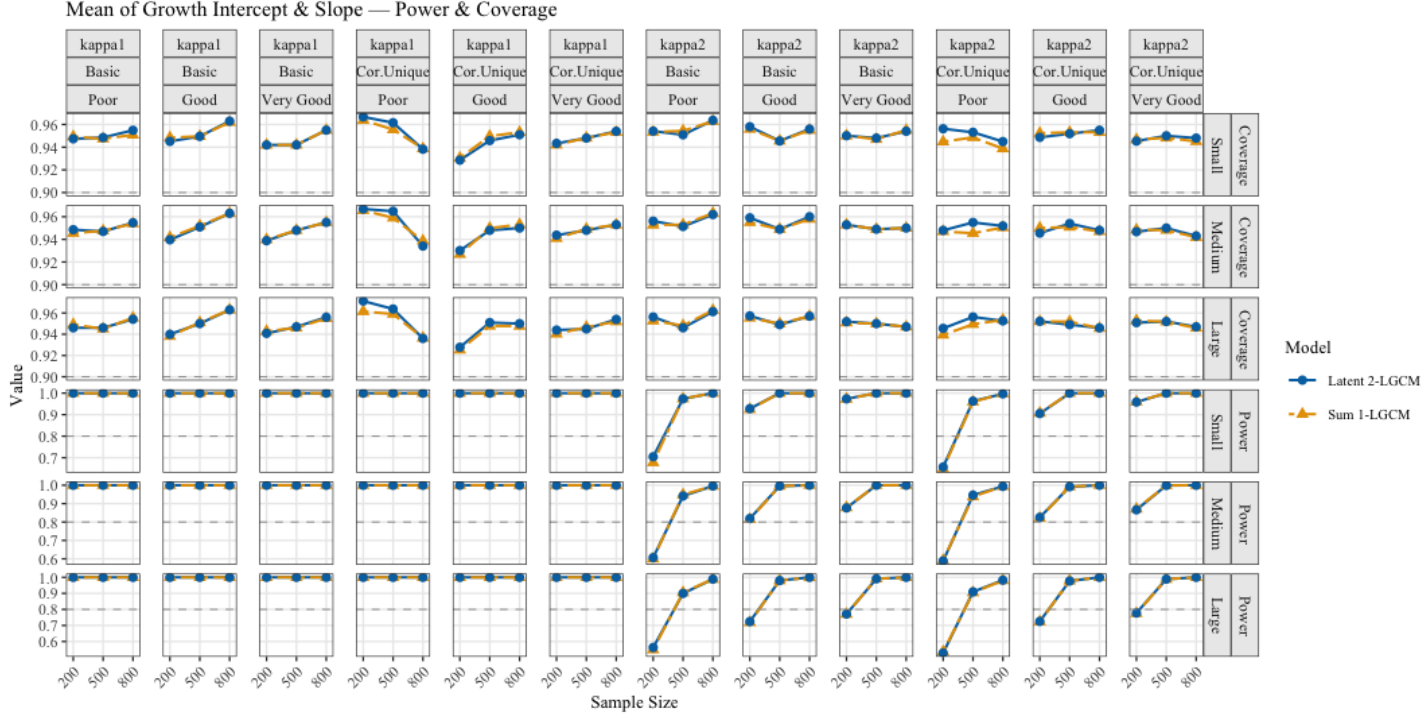


Figure 9. Power and 95% Coverage Rates on Means of Growth Intercept and Slope

Note. Basic = data and model without correlated uniqueness (residual covariance); Cor.Unique = data and model with correlated uniqueness. Poor/Good/Very Good reflect measurement quality (varying via residual variances). Small/Medium/Large denote growth slope variance magnitude respectively.

conditions were acceptable, generally exceeding 90%. Similarly, there was no meaningful difference in statistical power for the means of the growth intercept and slope between the two models (Figure 9). Finally, for admissible versus inadmissible solutions (Figure 10), when fitting the basic model without correlated residuals, the admissible solution rate was lower for the latent variable 2-LGCM than 1-LGCM using composite scores, as expected.

However, when correlated residuals were included and modeled, the admissible solution rate for 2-LGCM remained largely unchanged, whereas 1-LGCM using sum scores showed a substantial drop in admissible solutions, even lower than 2-LGCM using latent variables.

5.4 Discussion

The simulation provides a coherent answer to RQ1.1 and RQ1.2. Without correlated uniqueness (CU) and under ideal condition of strict longitudinal measurement invariance,

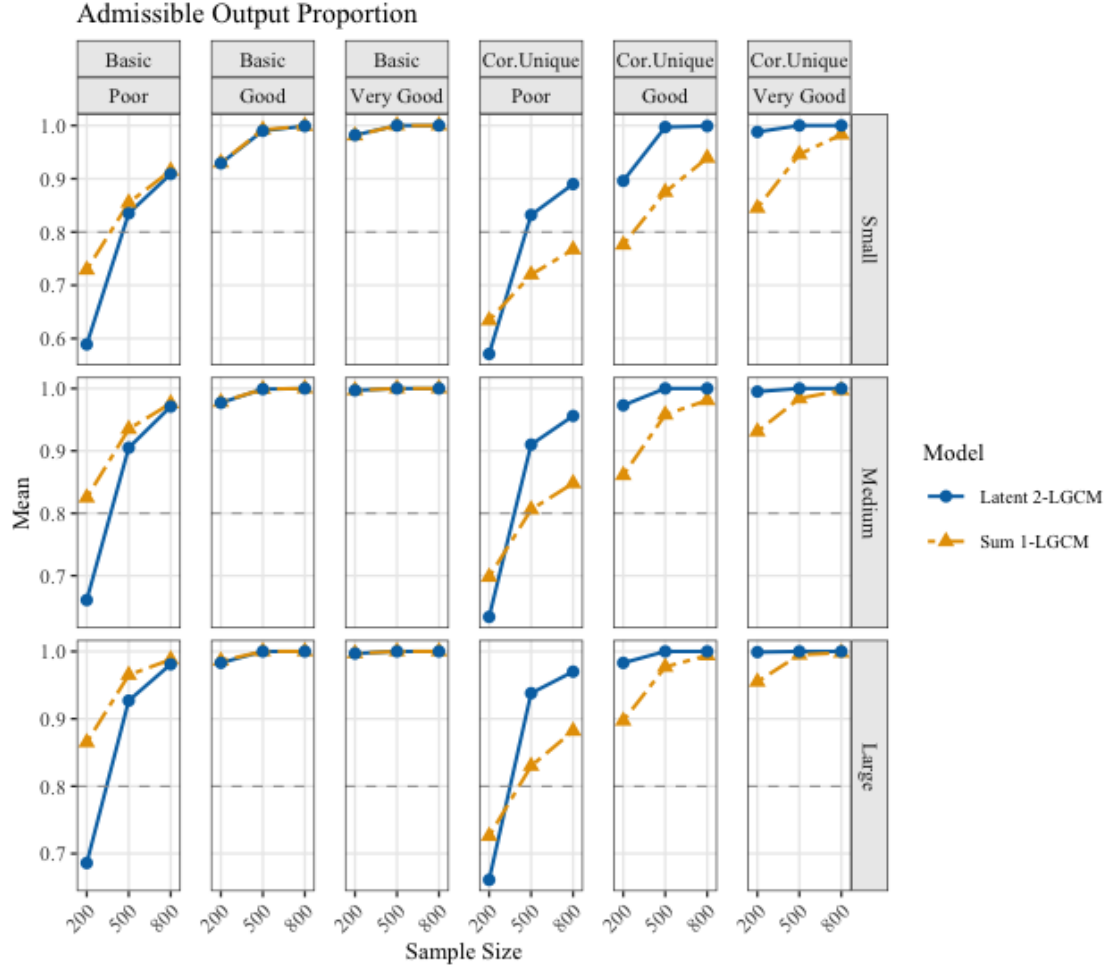


Figure 10. Proportion of Admissible Results

Note. Basic = data and model without correlated uniqueness (residual covariance); Cor.Unique = data and model with correlated uniqueness. Poor/Good/Very Good reflect measurement quality (varying via residual variances). Small/Medium/Large denote growth slope variance magnitude respectively.

1-LGCM and 2-LGCM produced essentially identical estimates of the growth means, variances, and covariance. Once CU was introduced, however, the two models diverged. Specifically, CU is a relationship in the residual component, so it leaves the mean trajectory untouched and the growth means κ_I and κ_S remain robust under both models regardless of the existence of CU. In the covariance structure, however, 1-LGCM has no parameter for residual covariance, so it has no choice but to absorb the shared residual variation into the growth factor variances. For the current simulation conditions, we found that the growth factor variances were inflated while the intercept–slope covariance was deflated, which seems a plausible pattern given that the total variance of the sum score is fixed.

Another important finding concerns admissibility. Once CU is properly modeled, retaining the 2-LGCM is advantageous despite its greater complexity relative to the 1-LGCM. The 1-LGCM has nowhere to absorb the residual misspecification (i.e., unmodeled CU), and forcing this absorption into a model without a measurement component appears to induce instability in estimation. Marsh and Hau (1996) previously reported that including CU did not adversely affect convergence rates in their simulations as well. Our results even indicate that omitting CU produces a substantial adverse effect on convergence. In short, while effect-coding identification aligns the two models on the mean structure, it does not keep their covariance structure intact against the presence of CU. Consequently, the simpler sum-score model proves not to be a necessarily safer alternative.

A major limitation of Study 1 is its reliance on ideal measurement conditions, because strict measurement invariance is rarely achievable in applied settings. Whereas Study 1 isolated one source of misspecification in the measurement part, namely the omission of CU, Study 2 turns to a different and broader source: the violation of longitudinal measurement invariance. The findings and limitations of Study 1 carry two direct implications for how Study 2 should be designed and interpreted. First, because the mean structure remained robust under misspecification of the residual component in Study 1, Study 2 will examine how the mean-related growth parameters are affected when factor loadings or item

intercepts are manipulated, both of which directly influence the expected sum score as shown in equation 12. Second, as shifting our interest towards the measurement non-invariance (MNI) manipulation, a scale-free metric needs to be used as a major tool for model comparison. As noted in Section 3.3 and Appendix A, under the MNI condition, scale equivalence between 1-LGCM and 2-LGCM no longer holds even after effect-coding adjustment, making direct comparisons of estimates (e.g., relative bias) impossible.

Chapter 6 Measurement Non-Invariance as a Threat to Growth Inference: Study 2

6.1 Purpose and Research Questions

Study 1 demonstrated that under measurement invariance (MI), the mean structure of growth parameters remains robust in the sum-score 1-LGCM, even when correlated uniqueness is omitted from the analysis model. This finding, however, rests on an assumption that is not always attainable in practice: invariant factor loadings and item intercepts across all measurement occasions. When these parameters shift over time, measurement non-invariance (MNI), the sum score at each wave can no longer be linearly scaled to the latent factor using time-constant weights. Consequently, applying effect-coding to the 2-LGCM loses its utility for establishing metric equivalence. Additionally, model fit indices will be evaluated, as MNI is typically detected through comparative model fit using the likelihood ratio test or differences in approximate fit indices.

Study 2 investigates exactly this conflicts: how, where, and how much, do sum-score and latent-variable-based growth models diverge when longitudinal measurement invariance is violated. The specific research questions for Study 2 are:

- RQ2.1: How do longitudinal measurement non-invariance, combined with varying levels of measurement quality and sample size, affect the admissible results rate, accuracy, and statistical power of outcome measures from 1-LGCM and 2-LGCM?
- RQ2.2: How does the model fit of 1-LGCM and 2-LGCM behave under conditions of measurement non-invariance across different sample sizes and measurement qualities?

6.2 Method

6.2.1 Data-Generating Model

The same basic data-generating model as the Study 1 was used as well with $I = 5$ indicators measured at $T = 5$ equally spaced time points by a linear latent growth process. The growth factor means were set to $\kappa_I = 1$ and $\kappa_S = 0.125$, and the growth factor covariance matrix was

$$\Phi = \begin{pmatrix} \phi_{II} & \phi_{IS} \\ \phi_{IS} & \phi_{SS} \end{pmatrix} = \begin{pmatrix} 1.00 & 0.18 \\ 0.18 & 0.203 \end{pmatrix},$$

yielding $\text{cor}(I, S) = 0.40$. In Study 2, we did not manipulate any of the growth parameters or rescale the population values using the effect-coding approach, as there is no way to align the 1-LGCM and 2-LGCM onto the same metric under MNI conditions. For the analysis model, we applied the marker-variable identification method by fixing the reference item's factor loadings to $\lambda = 1$, and their intercepts to $\tau = 1$. Monte Carlo replications were conducted by 2,000 times for each cell.

6.2.2 Simulation Conditions

Measurement Non-Invariance (MNI) MNI was defined by two sub-factors: the location of the non-invariance parameter and the severity of the MNI violation. Location had three levels: none, intercept (τ), and loading (λ). The manipulation of residual variance was not considered in the current study because invariant residual variances is often considered overly restrictive in applied research. In practice, establishing invariant factor loadings and intercepts is generally understood as sufficient for the valid comparison of latent means. MNI severity had three levels: none, weak, and strong. The weak condition introduced non-invariance in 1 of the 5 items (item 5), and the strong condition in 3 items (items 3, 4, and 5).

In both weak and strong MNI conditions, the parameters drifted over time following a

quadratic function (nonlinear), so that violation was absent at $t = 1$ and increased its maximum at $t = 5$ monotonically. To simulate this, a time-dependent magnitude was added to the baseline intercepts or factor loadings of the affected items. At the final wave, the weak condition applied a maximum additive shift of 0.25 for intercepts (δ_τ) or 0.15 for factor loadings (δ_λ). In contrast, the strong condition applied a larger maximum shift of 0.50 for intercepts or 0.30 for factor loadings across the three affected items. For example, under the strong loading MNI condition, the factor loading of each affected item at the final wave would increase by 0.30.

Size of Residual Variance To control the measurement quality at $t = 1$, residual variances (θ_i) were calibrated to yield specific target standardized loadings. Two conditions were specified: a 'very good' measurement quality level ($R^2_{\text{target}} = .49$; standardized loading = .70) and a 'poor' level ($R^2_{\text{target}} = .09$; standardized loading = .30). These residual variances were determined using the formula $\theta_i = \lambda^2 \cdot \text{Var}(\eta_1) \cdot (1/R^2_{\text{target}} - 1)$ relying on $\text{Var}(\eta_1)$ as the manipulation was specifically anchored at $t = 1$.

Correlated Uniqueness (ρ) The residual covariance structure for each indicator across time was also modeled as a first-order autoregressive pattern with adjacent-wave correlation ρ . Two levels were used: $\rho = 0$ (no correlated uniqueness) and $\rho = 0.15$ (small to moderate correlated uniqueness). For the 2-LGCM, adjacent-wave residual covariances for each item were freely estimated when $\rho > 0$; for the 1-LGCM, adjacent-wave covariances of the composite score were likewise freely estimated.

Sample Size (N) Three sample sizes were examined: $N = 200$, 500, and 800. These values span the range commonly encountered in longitudinal developmental and educational research, from a small but not uncommon panel ($N = 200$) to a moderately large sample ($N = 800$).

6.2.3 Evaluation Measures: Scale-Free

Correlation of Growth Intercept and Slope The correlation estimates between latent growth intercept and slope are evaluated to compare how well 1-LGCM and 2-LGCM recover the true population. Unlike other parameter estimates (e.g., variance of growth intercept), correlation is a standardized quantity where the covariance (ϕ_{IS}) is scaled by the standard deviations ($\sqrt{\phi_{II}}, \sqrt{\phi_{SS}}$). In other words, it provides information on how initial level is related to the rate of change regardless of scale. Thus, we consider it as an optimal measures for the comparison of 1-LGCM and 2-LGCM where the units are not the same by nature. The true population of correlation of growth slope and intercept was set as 0.4.

Standardized Mean Rate of Change (d_s) It is the mean growth slope standardized by the slope standard deviation ($\kappa_2/\sqrt{\phi_{22}}$), which shows an average rate of change within the variance of individual rates as Cohen's d . Since numerator and denominator are both expressed in the same within-model slope metric, the index is scale-free internally, which means that even when models operate on slope metrics that differ in absolute units, each model's d_s remains comparable on whether they agree about standardized magnitude of mean change or not. The generating population value for d_s was 0.278.

Model Fit Indices Considering that measurement invariance was evaluated by comparing model fits between models, we also included the model fit indices to inspect whether MNI is signaled out by improper model fit results. For the model fit indices, we included χ^2 p -value, Comparative Fit Index (CFI), Tucker-Lewis Index (TLI), Root Mean Square Error of Approximation (RMSEA), and Standardized Root Mean Square Residual (SRMR).

6.3 Results

Before examining how measurement non-invariance affect growth parameter estimation, it is essential to acknowledge a preliminary but consequential difference between the two modeling framework: their underlying metric. Because scale alignment between the 1-LGCM and 2-LGCM is not tenable under MNI conditions, we employed standard marker-variable identification for the 2-LGCM. Figures 11 and 12 present how the estimated means and variances of the growth factors differ solely due to this scale difference.

The most evident pattern is the systematic inflation of all sum-score 1-LGCM estimates to their 2-LGCM counterparts. This inflation is a direct consequence of aggregation: summing the $I = 5$ items rescales the construct by the sum of the factor loadings and sum of intercepts. Thus, raw parameter estimates from the two models cannot be compared on a one-to-one basis. Accordingly, all subsequent comparisons in this chapter rely on scale-free metrics (e.g., ρ_{IS} and standardized slope effect size $d_s = \kappa_2/\sqrt{\phi_{22}}$) that eliminate this artifact and isolate genuine divergence from model misspecification.

Figure 13 presents the correlation between growth factors (ρ_{IS}) and the standardized mean rate of change (d_s). Under the measurement invariance condition ('none'), most models converged toward the population value as the sample size increased. However, when MNI was introduced, there was a differential discrepancy between models depending on the type of MNI. Under intercept MNI, d_s was larger in the sum-score 1-LGCM, while ρ_{IS} was slightly underestimated if correlated uniqueness existed. On the other hand, when MNI was present in the factor loadings, both ρ_{IS} and d_s were overestimated in the sum-score 1-LGCM. In either MNI location condition, the divergences were greater with severe MNI, poor measurement quality, and the presence of correlated uniqueness.

Model fit indices of the sum-score 1-LGCM and the latent variable 2-LGCM were examined under the influence of MNI. Figure 14 summarizes the mean χ^2 p -values and RMSEAs. On average, under MNI condition, the χ^2 p -values became smaller in the

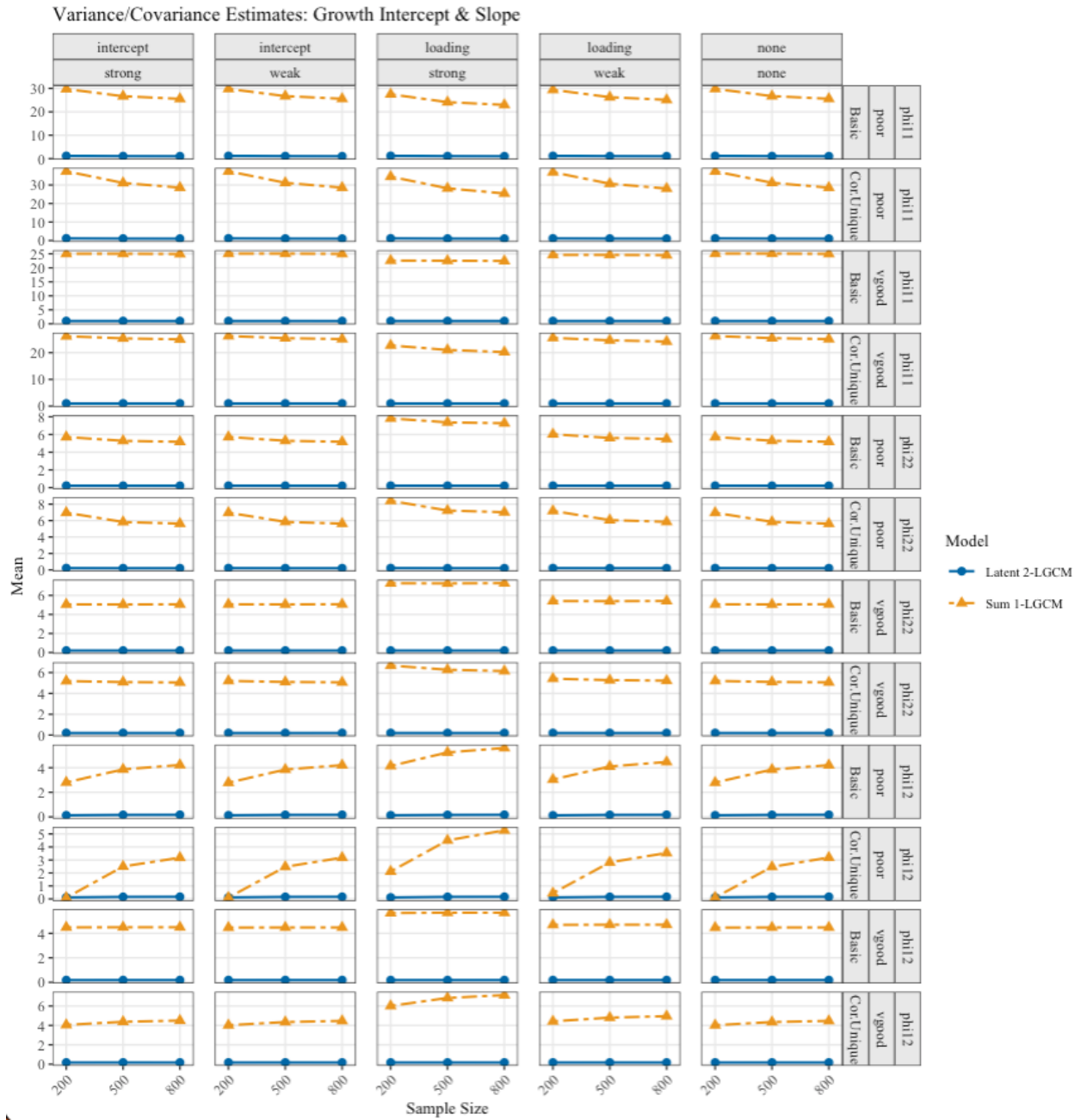


Figure 11. Mean Estimates of Growth Factor Variance and Covariance
Note. Basic = data and model without correlated uniqueness; Cor.Unique = with correlated uniqueness. Poor/Very Good reflect measurement quality (varying target loading).

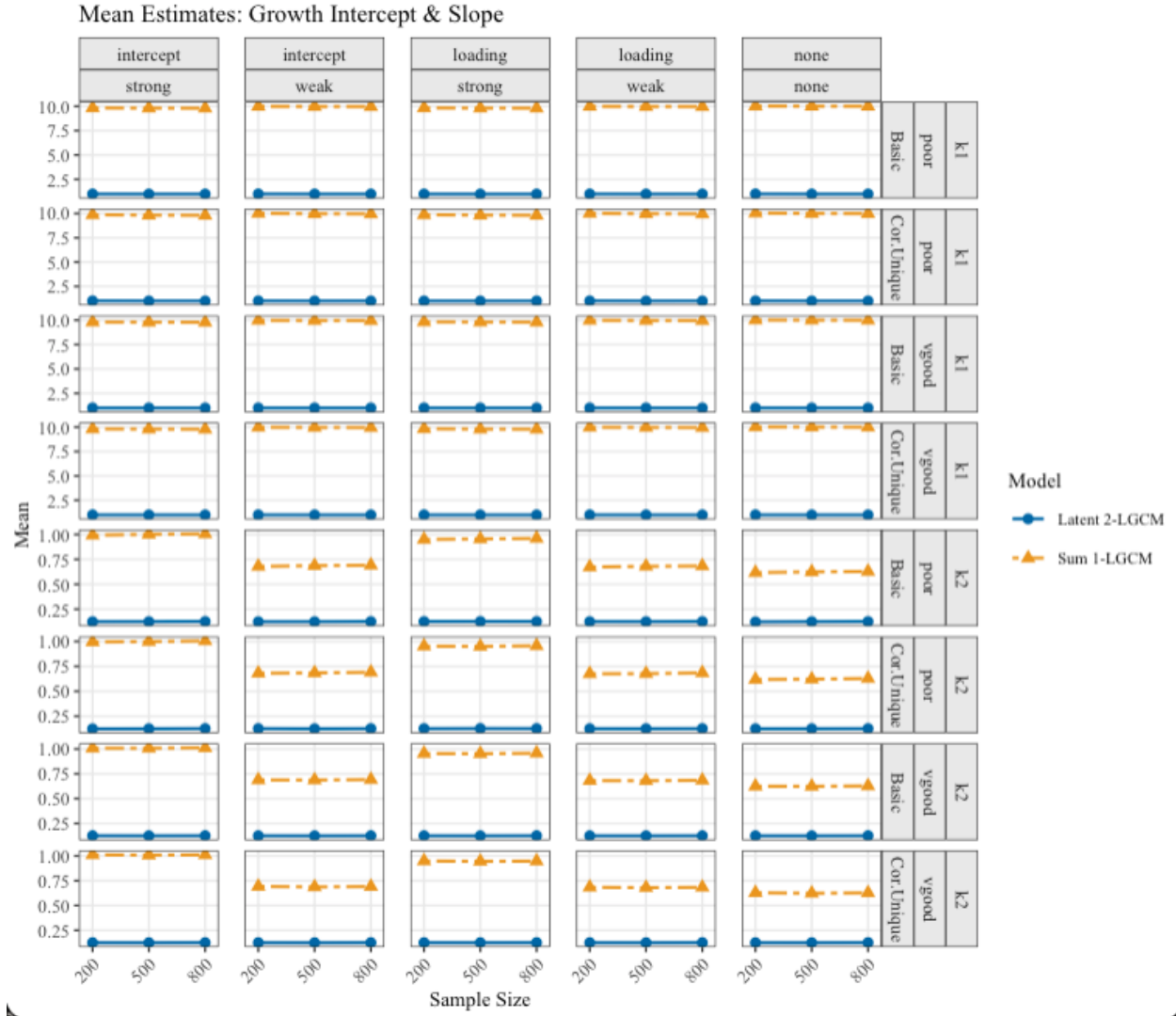


Figure 12. Mean Estimates of Growth Factor Means

Note. Basic = data and model without correlated uniqueness; Cor.Unique = with correlated uniqueness. Poor/Very Good reflect measurement quality (varying target loading).

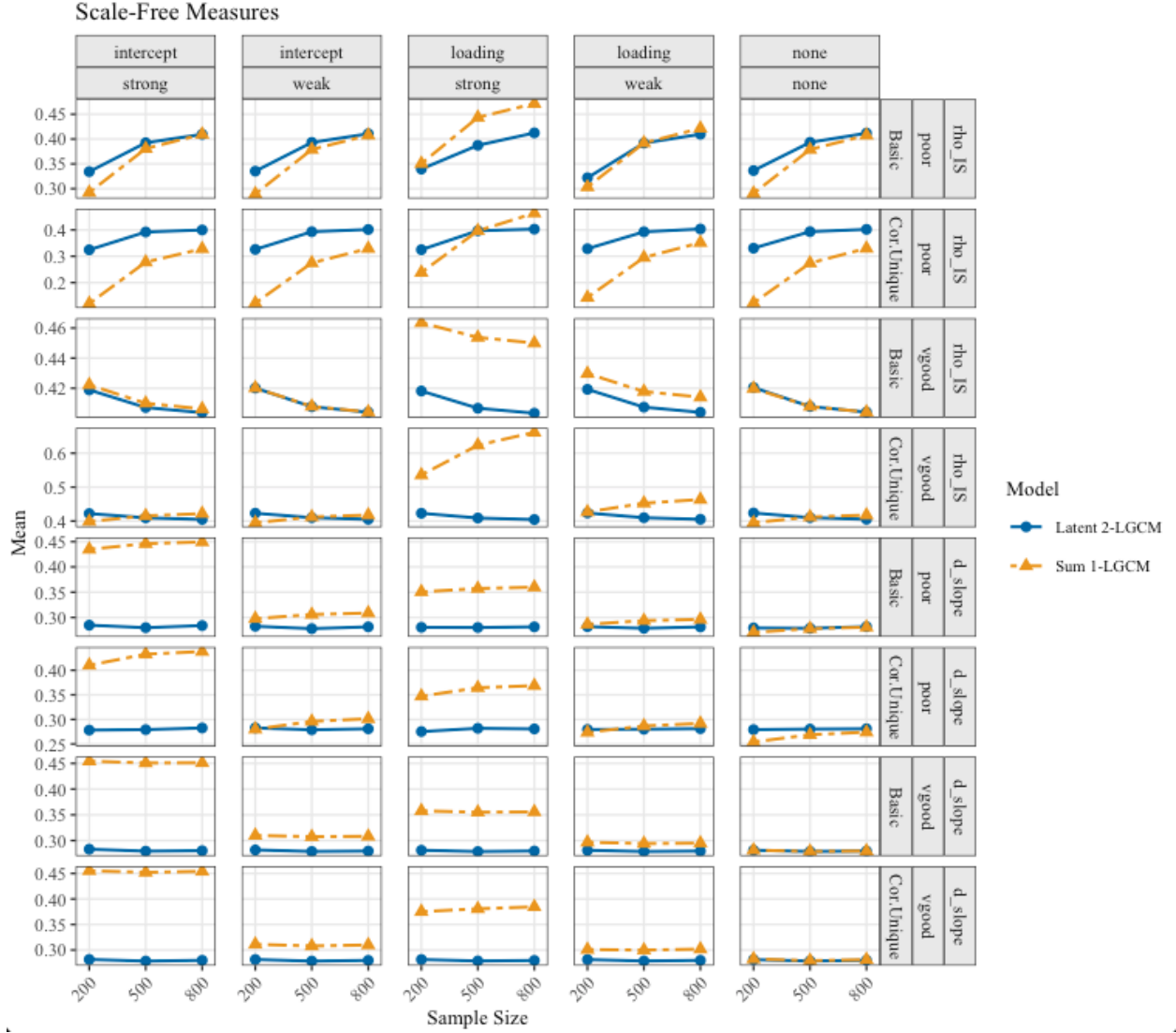


Figure 13. Mean Estimates of Scale-Free Measures

Note. Basic = data and model without correlated uniqueness; Cor.Unique = with correlated uniqueness. Poor/Very Good reflect measurement quality (varying target loading). d_s = standardized mean rate of change. ρ_{IS} = correlation between growth intercept and slope.

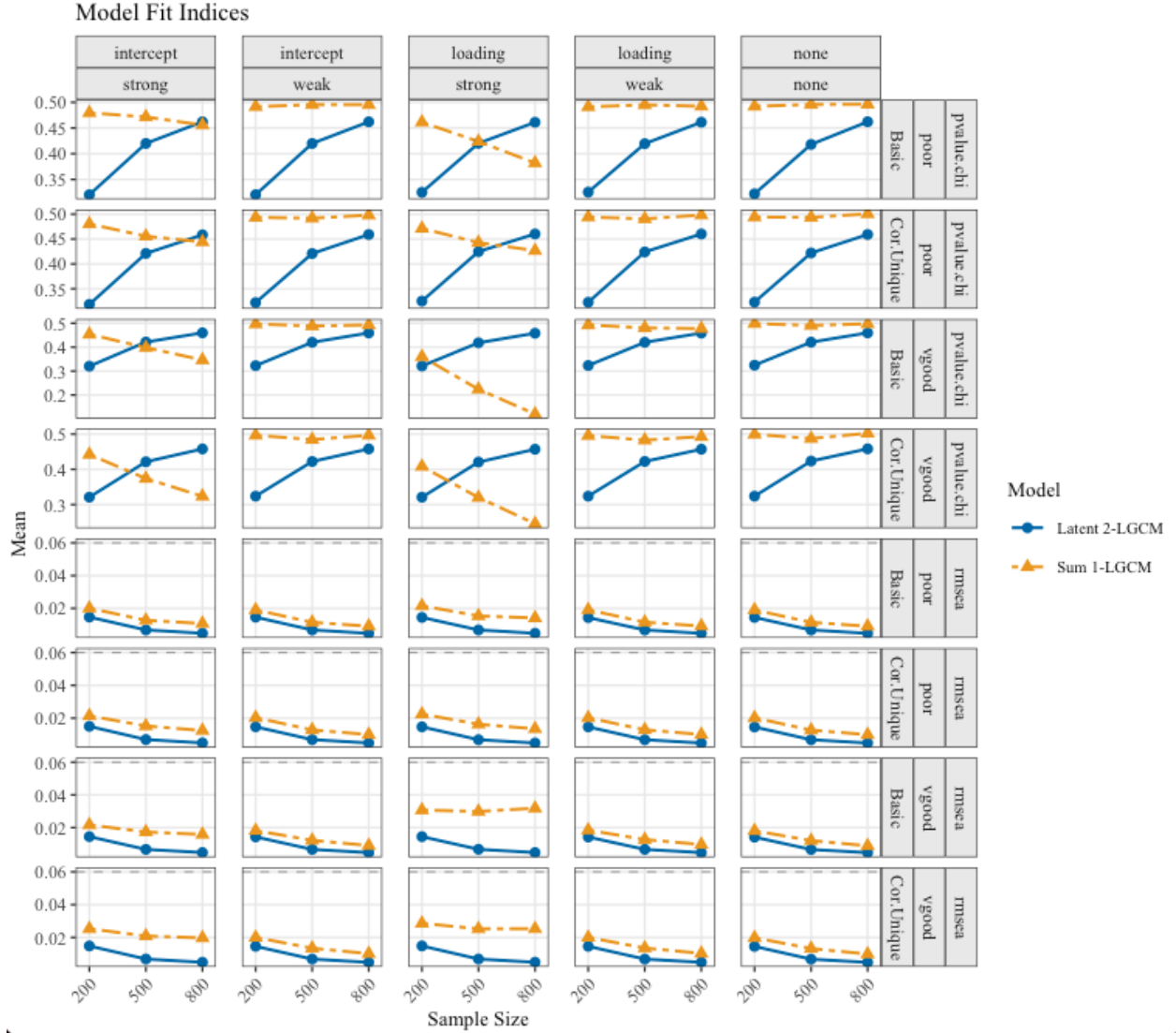


Figure 14. Chi-square P-value and RMSEA

Note. Basic = data and model without correlated uniqueness; Cor.Unique = with correlated uniqueness. Poor/Very Good reflect measurement quality (varying target loading).

sum-score 1-LGCM than in the 2-LGCM when sample size increases, leading to higher rejection rates, and the RMSEA was also slightly larger. This difference was most evident when there was strong MNI in the factor loadings.

In contrast, CFI, TLI, and SRMR showed different results, as illustrated in Figure 15. Across all conditions, the CFI of the sum-score 1-LGCM were higher than those of the latent 2-LGCM, while its SRMR was smaller and TLI performed similarly. Thus, even where parameter bias was present via MNI, these three approximate fit indices indicated

that the sum-score 1-LGCM fit the data well. In sum, there is a difference in diagnostic sensitivity among the fit indices; χ^2 and RMSEA could only provide a limited signal regarding misspecification, while CFI, TLI, and SRMR could not detect it.

Regarding the rate of admissible results shown in Figure 16, the sum-score 1-LGCM demonstrated a drastic drop in admissibility under the factor-loading MNI condition. Overall, the reversed pattern observed when correlated uniqueness is present remains consistent with the results from Study 1.

6.4 Discussion

Study 2 examined how measurement non-invariance (MNI) can yield diverging results between the sum-score 1-LGCM and the latent variable 2-LGCM. The relationship between the sum-score and the generating first-order latent factor ($\eta_t^{(gen)}$) is expressed as $S_t = c_t + a_t \eta_t^{(gen)} + \sum \varepsilon_{it}$ previously in equation (14). When measurement invariance holds (i.e., $a_t = a$, $c_t = c$), the expected mean and variance of the sum-score become a linear transformation of the generating first-order latent factor, observed empirically via simulation under measurement invariance and no omitted residual covariance condition. However, this equivalence is violated in the presence of MNI, and even scale-free measures showed diverging results between the models. This implies that the sum-score 1-LGCM is not merely a model on a different scale, but is only safely interchangeable under measurement invariance.

When intercept measurement non-invariance increased monotonically, d_s was inflated. This indicates that the expected sum scores increase more over time due to the increasing sum of the intercepts, producing a larger estimated rate of change (κ_2). Because this increasing intercept drift does not confound the variance-covariance component, the ratio ρ_{IS} does not appear to be affected, unlike the standardized mean rate of change (d_s). Thus, intercept MNI appears to preserve the individual difference structure, even though the shape of the mean trajectory may be biased.

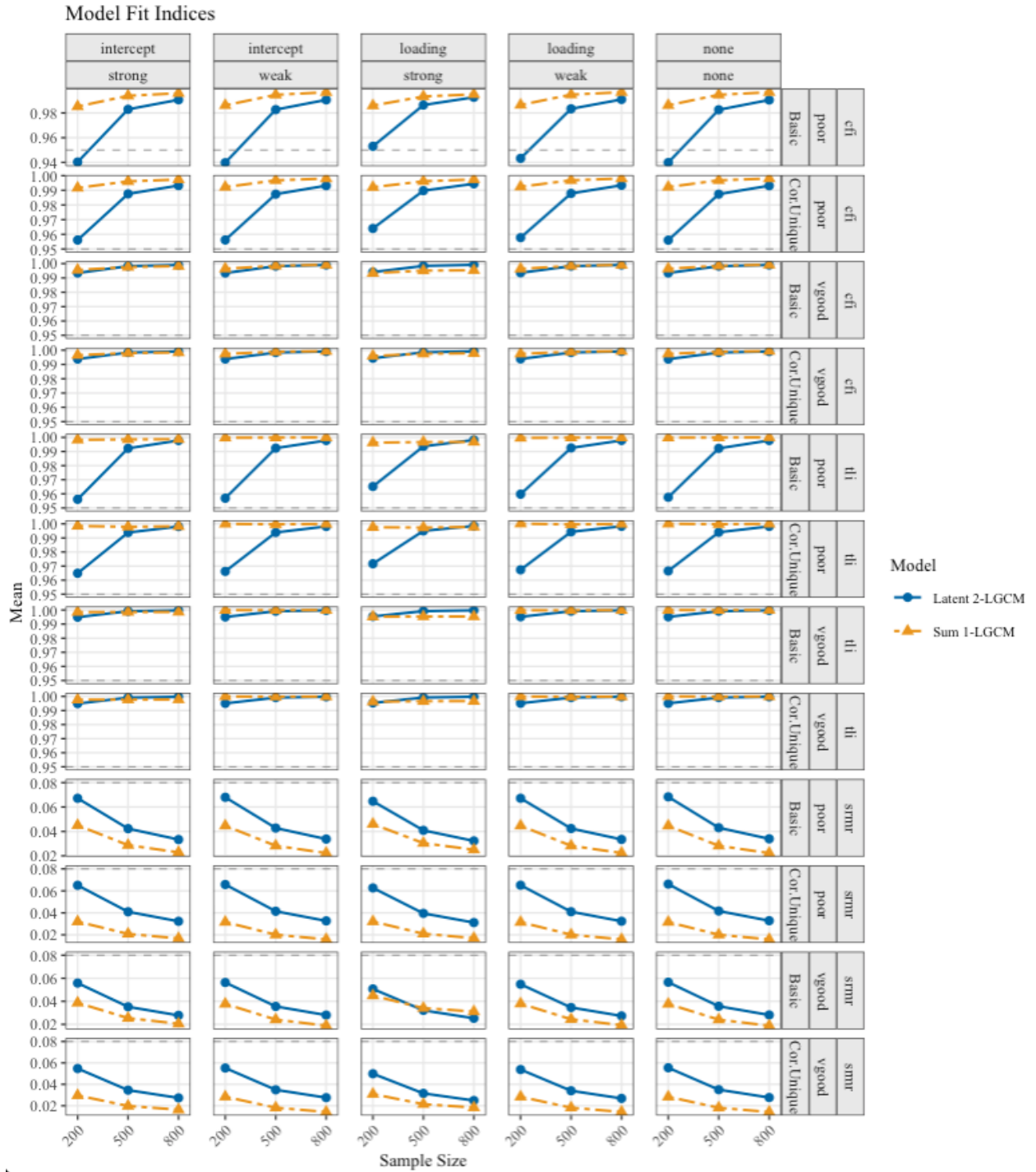


Figure 15. CFI, TLI, and RMSEA

Note. Basic = data and model without correlated uniqueness; Cor.Unique = with correlated uniqueness. Poor/Very Good reflect measurement quality (varying target loading).

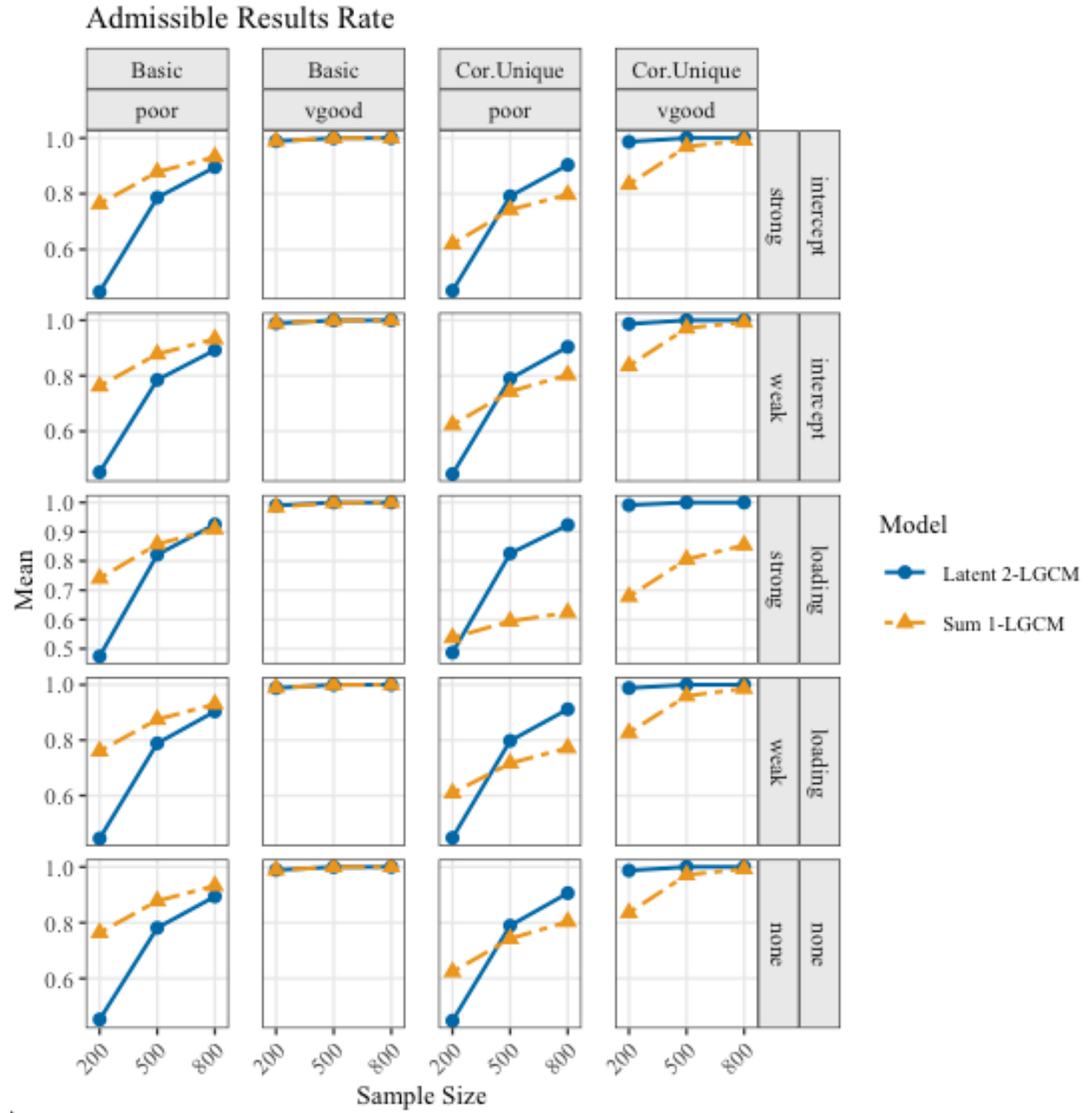


Figure 16. Proportion of Admissible Results

Note. Basic = data and model without correlated uniqueness; Cor.Unique = with correlated uniqueness. Poor/Very Good reflect measurement quality (varying target loading).

Conversely, monotonically increasing MNI in factor loadings caused ρ_{IS} to be underestimated in the sum-score 1-LGCM. As the sum of the factor loadings (a_t) increases over time, the variance of the sum-score increases drastically as it reaches the last time point, given that $Var(S_t) = a_t^2 Var(\eta_t^{(gen)}) + \dots$ following the equation (14). Compared to MNI in the intercepts, factor loading MNI distorts the individual difference structure more than the mean trajectory.

A major limitation of the current study is that we only allowed limited conditions regarding the size of growth parameters, lag-1 autocorrelation with adjacent time points, and monotonically increasing measurement parameter drifts. For future studies, more varied conditions that can better represent real applied settings should be examined. For example, an interactive tool could be developed to illustrate how much drift in measurement parameters can distort the trajectory and growth parameter estimates depending on the type of scores used. Also, based on the fingerprints of measurement non-invariance observed in the sum-score 1-LGCM, a classifier to detect potential measurement non-invariance could be developed using observed data.

Chapter 7 (Proposed) Pre-Fit Diagnostics for Detecting Measurement Non-Invariance: Study 3

7.1 Proposal for Study 3

Studies 1 and 2 demonstrated that omitted correlated uniqueness and longitudinal measurement non-invariance can systematically distort growth parameter estimates in the sum-score 1-LGCM. However, both studies were operated, given researchers know the location of these threats a priori in model specification. In applied settings, the only way to evaluate them is to fit a 2-LGCM, conduct invariance testing, and explicitly model residual covariance structure. This is precisely the procedure that the sum-score 1-LGCM was chosen to avoid. This creates a circular problem for applied researchers: to determine whether the simpler 1-LGCM is safe or not, they must first fit the more complex 2-LGCM, which requires large samples, often fails to converge, and demands SEM expertise. In addition, the approximate fit indices examined in Study 2 (CFI, TLI, and SRMR) failed to flag bias from MNI when using sum-score 1-LGCM, which provides limited diagnostic guidance.

Study 3 aims to address this gap by exploring whether supervised learning based classifiers, trained on observed item-level data, can signal the presence of MNI or CU before any growth model is fit. The candidate approach is to use the item-level statistics that capture cross-item heterogeneity in how item means and variances differ over time, largely motivated by the algebraic relationship between measurement parameters and the sum-score derived in equation (17) at Chapter 2.2. These features are then used as inputs to the supervised classifiers, including a logistic regression as a baseline and ensemble methods such as random forest and gradient boosting. For comparison, conventional SEM-based diagnostics, such as sequential LRT and approximate fit index differences, will be evaluated.

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Appendix A: Scale Alignment of Growth Parameters Under Effect Coding

Building on the metric equivalence in sum-score from 1-LGCM and effect-coded latent variable from 2-LGCM in Section 3.2, this page examines whether the growth factor mapping between the generating-model and the effect-coded model remains defined or not when measurement parameters are allowed to vary across time.

A.1 Under Measurement Invariance: Single Solution Exists

The latent variable from EC-fitted model ($\eta_t^{(ec)}$) can be written in terms of growth factor from generating model $\boldsymbol{\xi}^{(gen)}$ via two ways. From the factor-level transformation (24) and the generating growth model (9),

$$\eta_t^{(ec)} = a_t \eta_t^{(gen)} + c_t = a_t (\mathbf{b}_t' \boldsymbol{\xi}^{(gen)} + \zeta_t^{(gen)}) + c_t.$$

The coefficient on $\boldsymbol{\xi}^{(gen)}$ is $a_t \mathbf{b}_t'$. On the other hand, if the EC growth factors could be assumed to be a linear transformation of the generating growth factors, let

$$\boldsymbol{\xi}^{(ec)} = \mathbf{M} \boldsymbol{\xi}^{(gen)} + \mathbf{d}, \tag{28}$$

then the EC growth model gives

$$\eta_t^{(ec)} = \mathbf{b}_t' \boldsymbol{\xi}^{(ec)} + \zeta_t^{(ec)} = \mathbf{b}_t' (\mathbf{M} \boldsymbol{\xi}^{(gen)} + \mathbf{d}) + \zeta_t^{(ec)}.$$

Then, the coefficient on $\boldsymbol{\xi}^{(gen)}$ is $\mathbf{b}_t' \mathbf{M}$. Because both expressions represent the same $\eta_t^{(ec)}$ and the growth factors $\boldsymbol{\xi}^{(gen)}$ are independent of the disturbances $\zeta_t^{(gen)}$, equating the coefficients on $\boldsymbol{\xi}^{(gen)}$ across the two sides yields, in matrix form

$$\mathbf{B} \mathbf{M} = \mathbf{A} \mathbf{B}, \tag{29}$$

where $\mathbf{A} = \text{diag}(a_1, \dots, a_T)$. The same logic applied to the location parameters yields $\mathbf{B}\mathbf{d} = \mathbf{c}$, and the disturbance terms give $\zeta_t^{(\text{ec})} = a_t \zeta_t^{(\text{gen})}$, although the latter is not of primary interest here.

Under measurement invariance, $\mathbf{\Lambda}_t = \mathbf{\Lambda}$ and $\boldsymbol{\tau}_t = \boldsymbol{\tau}$ for all t , so $a_t = a$ and $c_t = c(\text{constant})$. Then, the equation 29 becomes

$$\mathbf{B}\mathbf{M} = a\mathbf{B}, \quad (30)$$

which is satisfied by $\mathbf{M} = a\mathbf{I}_2$. Using the new definition of 28, then

$$\boldsymbol{\xi}^{(\text{ec})} = a\boldsymbol{\xi}^{(\text{gen})} + \mathbf{d},$$

Substituting this into the expected means of growth-factor ($\boldsymbol{\kappa}$) yields

$$E[\boldsymbol{\xi}^{(\text{ec})}] = \boldsymbol{\kappa}^{(\text{ec})} = a\boldsymbol{\kappa}^{(\text{gen})} + \mathbf{d} \quad (31)$$

The same scalar a also governs the variance relationship as:

$$\text{Var}[\boldsymbol{\xi}^{(\text{ec})}] = \boldsymbol{\Phi}^{(\text{ec})} = a^2\boldsymbol{\Phi}^{(\text{gen})}. \quad (32)$$

Thus, using relationship from 31 and 32 with the scalar a by measurement none-invariance, we can back-map the growth parameters in an effect-coded scale.

A.2 Under Measurement Non-Invariance: No Single Solution Exists

Under measurement non-invariance, the scaling factors $a_1, a_2, \dots, a_5$ vary across time points (i.e., a_t are not all equal). To understand why the mapping equation $\mathbf{B}\mathbf{M} = \mathbf{A}\mathbf{B}$

generically has no single solution in this scenario, we first define the components:

$$\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ \vdots & \vdots \\ 1 & T-1 \end{pmatrix}, \quad \mathbf{M} = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix},$$

where T is 5 in our example at Study 1 and 2. Here, $\mathbf{M}$ is a 2×2 matrix that links the generating growth factors to the EC-fitted growth factors. For the equality $\mathbf{B}\mathbf{M} = \mathbf{A}\mathbf{B}$ to hold for every time point, the t -th row of both sides must be equal. Since $\mathbf{A} = \text{diag}(a_1, \dots, a_T)$ with full constants, evaluating row t yields the following system of equations:

$$m_{11} + m_{21}(t-1) = a_t, \quad (33)$$

$$m_{12} + m_{22}(t-1) = a_t(t-1). \quad (34)$$

Equation (33) shows that a_t must be linear in t . If a_t is nonlinear, m_{11} and m_{21} cannot exist. Even if a_t happens to be linear, equation (34) requires $a_t(t-1)$ to also be linear in t . The only way both a_t and $a_t(t-1)$ can be linear simultaneously is if the slope of a_t is zero, meaning a_t is constant across all time points. This implies measurement invariance, which contradicts the non-invariance assumption. Thus, no such a single matrix $\mathbf{M}$ solution exists when a_t varies over time, but only approximation can be used for proxy estimates.